```
!pip install pillow matplotlib
!pip install tensorflow-gpu==2.0.0
```

```
Requirement already satisfied: pillow in
/usr/local/lib/python3.7/dist-packages (7.1.2)
Requirement already satisfied: matplotlib in
/usr/local/lib/python3.7/dist-packages (3.2.2)
Requirement already satisfied: kiwisolver>=1.0.1 in
/usr/local/lib/python3.7/dist-packages (from matplotlib) (1.4.0)
Requirement already satisfied: pyparsing!=2.0.4,!=2.1.2,!
=2.1.6,>=2.0.1 in /usr/local/lib/python3.7/dist-packages (from
matplotlib) (3.0.7)
Requirement already satisfied: python-dateutil>=2.1 in
/usr/local/lib/python3.7/dist-packages (from matplotlib) (2.8.2)
Requirement already satisfied: numpy>=1.11 in
/usr/local/lib/python3.7/dist-packages (from matplotlib) (1.21.5)
Requirement already satisfied: cycler>=0.10 in
/usr/local/lib/python3.7/dist-packages (from matplotlib) (0.11.0)
Requirement already satisfied: typing-extensions in
/usr/local/lib/python3.7/dist-packages (from kiwisolver>=1.0.1-
>matplotlib) (3.10.0.2)
Requirement already satisfied: six>=1.5 in
/usr/local/lib/python3.7/dist-packages (from python-dateutil>=2.1-
>matplotlib) (1.15.0)
Collecting tensorflow-gpu==2.0.0
  Downloading tensorflow_gpu-2.0.0-cp37-cp37m-manylinux2010_x86_64.whl
(380.8 MB)
ent already satisfied: opt-einsum>=2.3.2 in
/usr/local/lib/python3.7/dist-packages (from tensorflow-gpu==2.0.0)
(3.3.0)
Requirement already satisfied: wrapt>=1.11.1 in
/usr/local/lib/python3.7/dist-packages (from tensorflow-gpu==2.0.0)
(1.14.0)
Requirement already satisfied: protobuf>=3.6.1 in
/usr/local/lib/python3.7/dist-packages (from tensorflow-gpu==2.0.0)
(3.17.3)
Collecting gast==0.2.2
  Downloading gast-0.2.2.tar.gz (10 kB)
Requirement already satisfied: numpy<2.0,>=1.16.0 in
/usr/local/lib/python3.7/dist-packages (from tensorflow-gpu==2.0.0)
(1.21.5)
Collecting tensorboard<2.1.0,>=2.0.0
  Downloading tensorboard-2.0.2-py3-none-any.whl (3.8 MB)
ent already satisfied: six>=1.10.0 in /usr/local/lib/python3.7/dist-
packages (from tensorflow-gpu==2.0.0) (1.15.0)
Requirement already satisfied: astor>=0.6.0 in
/usr/local/lib/python3.7/dist-packages (from tensorflow-gpu==2.0.0)
(0.8.1)
Requirement already satisfied: google-pasta>=0.1.6 in
/usr/local/lib/python3.7/dist-packages (from tensorflow-gpu==2.0.0)
```

```
(0.2.0)
Requirement already satisfied: absl-py>=0.7.0 in
/usr/local/lib/python3.7/dist-packages (from tensorflow-gpu==2.0.0)
(1.0.0)
Collecting keras-applications>=1.0.8
  Downloading Keras_Applications-1.0.8-py3-none-any.whl (50 kB)
ent already satisfied: grpcio>=1.8.6 in /usr/local/lib/python3.7/dist-
packages (from tensorflow-gpu==2.0.0) (1.44.0)
Collecting tensorflow-estimator<2.1.0,>=2.0.0
  Downloading tensorflow_estimator-2.0.1-py2.py3-none-any.whl (449 kB)
ent already satisfied: termcolor>=1.1.0 in
/usr/local/lib/python3.7/dist-packages (from tensorflow-gpu==2.0.0)
(1.1.0)
Requirement already satisfied: wheel>=0.26 in
/usr/local/lib/python3.7/dist-packages (from tensorflow-gpu==2.0.0)
(0.37.1)
Requirement already satisfied: keras-preprocessing>=1.0.5 in
/usr/local/lib/python3.7/dist-packages (from tensorflow-gpu==2.0.0)
(1.1.2)
Requirement already satisfied: h5py in /usr/local/lib/python3.7/dist-
packages (from keras-applications>=1.0.8->tensorflow-gpu==2.0.0)
(3.1.0)
Requirement already satisfied: google-auth-oauthlib<0.5,>=0.4.1 in
/usr/local/lib/python3.7/dist-packages (from
tensorboard<2.1.0,>=2.0.0->tensorflow-gpu==2.0.0) (0.4.6)
Requirement already satisfied: werkzeug>=0.11.15 in
/usr/local/lib/python3.7/dist-packages (from
tensorboard<2.1.0,>=2.0.0->tensorflow-gpu==2.0.0) (1.0.1)
Requirement already satisfied: google-auth<2,>=1.6.3 in
/usr/local/lib/python3.7/dist-packages (from
tensorboard<2.1.0,>=2.0.0->tensorflow-gpu==2.0.0) (1.35.0)
Requirement already satisfied: setuptools>=41.0.0 in
/usr/local/lib/python3.7/dist-packages (from
tensorboard<2.1.0,>=2.0.0->tensorflow-gpu==2.0.0) (57.4.0)
Requirement already satisfied: requests<3,>=2.21.0 in
/usr/local/lib/python3.7/dist-packages (from
tensorboard<2.1.0,>=2.0.0->tensorflow-gpu==2.0.0) (2.23.0)
Requirement already satisfied: markdown>=2.6.8 in
/usr/local/lib/python3.7/dist-packages (from
tensorboard<2.1.0,>=2.0.0->tensorflow-gpu==2.0.0) (3.3.6)
Requirement already satisfied: pyasn1-modules>=0.2.1 in
/usr/local/lib/python3.7/dist-packages (from google-auth<2,>=1.6.3-
>tensorboard<2.1.0,>=2.0.0->tensorflow-gpu==2.0.0) (0.2.8)
Requirement already satisfied: rsa<5,>=3.1.4 in
/usr/local/lib/python3.7/dist-packages (from google-auth<2,>=1.6.3-
>tensorboard<2.1.0,>=2.0.0->tensorflow-gpu==2.0.0) (4.8)
Requirement already satisfied: cachetools<5.0,>=2.0.0 in
/usr/local/lib/python3.7/dist-packages (from google-auth<2,>=1.6.3-
>tensorboard<2.1.0,>=2.0.0->tensorflow-gpu==2.0.0) (4.2.4)
Requirement already satisfied: requests-oauthlib>=0.7.0 in
```

```
/usr/local/lib/python3.7/dist-packages (from google-auth-
oauthlib<0.5,>=0.4.1->tensorboard<2.1.0,>=2.0.0->tensorflow-
gpu==2.0.0) (1.3.1)
Requirement already satisfied: importlib-metadata>=4.4 in
/usr/local/lib/python3.7/dist-packages (from markdown>=2.6.8-
>tensorboard<2.1.0,>=2.0.0->tensorflow-gpu==2.0.0) (4.11.3)
Requirement already satisfied: zipp>=0.5 in
/usr/local/lib/python3.7/dist-packages (from importlib-metadata>=4.4-
>markdown>=2.6.8->tensorboard<2.1.0,>=2.0.0->tensorflow-gpu==2.0.0)
(3.7.0)
Requirement already satisfied: typing-extensions>=3.6.4 in
/usr/local/lib/python3.7/dist-packages (from importlib-metadata>=4.4-
>markdown>=2.6.8->tensorboard<2.1.0,>=2.0.0->tensorflow-gpu==2.0.0)
(3.10.0.2)
Requirement already satisfied: pyasn1<0.5.0,>=0.4.6 in
/usr/local/lib/python3.7/dist-packages (from pyasn1-modules>=0.2.1-
>google-auth<2,>=1.6.3->tensorboard<2.1.0,>=2.0.0->tensorflow-
gpu==2.0.0) (0.4.8)
Requirement already satisfied: idna<3,>=2.5 in
/usr/local/lib/python3.7/dist-packages (from requests<3,>=2.21.0-
>tensorboard<2.1.0,>=2.0.0->tensorflow-gpu==2.0.0) (2.10)
Requirement already satisfied: chardet<4,>=3.0.2 in
/usr/local/lib/python3.7/dist-packages (from requests<3,>=2.21.0-
>tensorboard<2.1.0,>=2.0.0->tensorflow-gpu==2.0.0) (3.0.4)
Requirement already satisfied: urllib3!=1.25.0,!=1.25.1,<1.26,>=1.21.1
in /usr/local/lib/python3.7/dist-packages (from requests<3,>=2.21.0-
>tensorboard<2.1.0,>=2.0.0->tensorflow-gpu==2.0.0) (1.24.3)
Requirement already satisfied: certifi>=2017.4.17 in
/usr/local/lib/python3.7/dist-packages (from requests<3,>=2.21.0-
>tensorboard<2.1.0,>=2.0.0->tensorflow-gpu==2.0.0) (2021.10.8)
Requirement already satisfied: oauthlib>=3.0.0 in
/usr/local/lib/python3.7/dist-packages (from requests-oauthlib>=0.7.0-
>google-auth-oauthlib<0.5,>=0.4.1->tensorboard<2.1.0,>=2.0.0-
>tensorflow-gpu==2.0.0) (3.2.0)
Requirement already satisfied: cached-property in
/usr/local/lib/python3.7/dist-packages (from h5py->keras-
applications>=1.0.8->tensorflow-gpu==2.0.0) (1.5.2)
Building wheels for collected packages: gast
  Building wheel for gast (setup.py) ... e=gast-0.2.2-py3-none-any.whl
size=7554
sha256=2266d1b77376816012086c478b0e315bed54f0df51ae980f0295e1253668a5e
d
  Stored in directory:
/root/.cache/pip/wheels/21/7f/02/420f32a803f7d0967b48dd823da3f558c5166
991bfd204eef3
Successfully built gast
Installing collected packages: tensorflow-estimator, tensorboard,
keras-applications, gast, tensorflow-gpu
  Attempting uninstall: tensorflow-estimator
    Found existing installation: tensorflow-estimator 2.8.0
```

```
    Uninstalling tensorflow-estimator-2.8.0:
      Successfully uninstalled tensorflow-estimator-2.8.0
  Attempting uninstall: tensorboard
    Found existing installation: tensorboard 2.8.0
    Uninstalling tensorboard-2.8.0:
      Successfully uninstalled tensorboard-2.8.0
  Attempting uninstall: gast
    Found existing installation: gast 0.5.3
    Uninstalling gast-0.5.3:
      Successfully uninstalled gast-0.5.3
ERROR: pip's dependency resolver does not currently take into account
all the packages that are installed. This behaviour is the source of
the following dependency conflicts.
tensorflow 2.8.0 requires tf-estimator-nightly==2.8.0.dev2021122109,
which is not installed.
tensorflow 2.8.0 requires tensorboard<2.9,>=2.8, but you have
tensorboard 2.0.2 which is incompatible.
tensorflow-probability 0.16.0 requires gast>=0.3.2, but you have gast
0.2.2 which is incompatible.
Successfully installed gast-0.2.2 keras-applications-1.0.8
tensorboard-2.0.2 tensorflow-estimator-2.0.1 tensorflow-gpu-2.0.0
```

```
!pip install 'h5py==2.10.0'
```

```
Collecting h5py==2.10.0
  Downloading h5py-2.10.0-cp37-cp37m-manylinux1_x86_64.whl (2.9 MB)
ent already satisfied: six in /usr/local/lib/python3.7/dist-packages
(from h5py==2.10.0) (1.15.0)
Requirement already satisfied: numpy>=1.7 in
/usr/local/lib/python3.7/dist-packages (from h5py==2.10.0) (1.21.5)
Installing collected packages: h5py
  Attempting uninstall: h5py
    Found existing installation: h5py 3.1.0
    Uninstalling h5py-3.1.0:
      Successfully uninstalled h5py-3.1.0
ERROR: pip's dependency resolver does not currently take into account
all the packages that are installed. This behaviour is the source of
the following dependency conflicts.
tensorflow 2.8.0 requires tf-estimator-nightly==2.8.0.dev2021122109,
which is not installed.
tensorflow 2.8.0 requires tensorboard<2.9,>=2.8, but you have
tensorboard 2.0.2 which is incompatible.
Successfully installed h5py-2.10.0
```

```
from google.colab import drive
drive.mount('/content/drive')
```

```
Mounted at /content/drive
```

```
# Copyright 2019 NVIDIA Corporation. All Rights Reserved.
#
```

```
# Licensed under the Apache License, Version 2.0 (the "License");
# you may not use this file except in compliance with the License.
# You may obtain a copy of the License at
#
#     http://www.apache.org/licenses/LICENSE-2.0
#
# Unless required by applicable law or agreed to in writing, software
# distributed under the License is distributed on an "AS IS" BASIS,
# WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or
implied.
# See the License for the specific language governing permissions and
# limitations under the License.
#
================================================================================
========
```

TF-TRT Inference from Keras Model with TensorFlow 2.0

Introduction

The NVIDIA TensorRT is a C++ library that facilitates high performance inference on NVIDIA graphics processing units (GPUs). TensorRT takes a trained network, which consists of a network definition and a set of trained parameters, and produces a highly optimized runtime engine which performs inference for that network.

TensorFlow™ integration with TensorRT™ (TF-TRT) optimizes and executes compatible subgraphs, allowing TensorFlow to execute the remaining graph. While you can still use TensorFlow's wide and flexible feature set, TensorRT will parse the model and apply optimizations to the portions of the graph wherever possible.

In this notebook, we demonstrate the process of creating a TF-TRT optimized model from a ResNet-50 Keras saved model.

Requirement

GPU

Before running this notebook, please set the Colab runtime environment to GPU via the menu *Runtime => Change runtime type => GPU*.

This demo will work on any NVIDIA GPU with CUDA cores, though for improved FP16 and INT8 inference, a Volta, Turing or newer generation GPU with Tensor cores is desired. On Google Colab, this normally means a T4 GPU. If you are assigned an older K80 GPU, another trial at another time might give you a T4 GPU.

```
!nvidia-smi
```

```
Fri Apr  8 16:26:03 2022
+-----------------------------------------------------------------------------
--------+
| NVIDIA-SMI 460.32.03    Driver Version: 460.32.03    CUDA Version:
11.2     |
|-------------------------------+----------------------
+----------------------+
| GPU  Name        Persistence-M| Bus-Id        Disp.A | Volatile
Uncorr. ECC |
| Fan  Temp  Perf  Pwr:Usage/Cap|         Memory-Usage | GPU-Util
Compute M. |
|                               |                      |
MIG M. |
|
===============================+======================+===============
=======|
|   0  Tesla T4            Off  | 00000000:00:04.0 Off |
0 |
| N/A   37C    P8     9W /  70W |      0MiB / 15109MiB |      0%
Default |
|                               |                      |
N/A |
+-------------------------------+----------------------
+----------------------+


+-----------------------------------------------------------------------------
--------+
| Processes:
|
|  GPU   GI   CI        PID   Type   Process name                  GPU
Memory |
|        ID   ID
Usage      |
|
=============================================================================
=======|
|  No running processes found
|
+-----------------------------------------------------------------------------
--------+
```

Install TensorFlow-GPU 2.0

```
import tensorflow as tf
print("Tensorflow version: ", tf.version.VERSION)
```

```
Tensorflow version:  2.0.0
```

Install TensorRT Runtime

```
%%bash
wget https://developer.download.nvidia.com/compute/machine-learning/repos/ubuntu1804/x86_64/nvidia-machine-learning-repo-ubuntu1804_1.0.0-1_amd64.deb

dpkg -i nvidia-machine-learning-repo-*.deb
apt-get update

sudo apt-get install libnvinfer5
```

```
Selecting previously unselected package nvidia-machine-learning-repo-ubuntu1804.
(Reading database ... 156210 files and directories currently installed.)
Preparing to unpack nvidia-machine-learning-repo-ubuntu1804_1.0.0-1_amd64.deb ...
Unpacking nvidia-machine-learning-repo-ubuntu1804 (1.0.0-1) ...
Setting up nvidia-machine-learning-repo-ubuntu1804 (1.0.0-1) ...
Get:1 https://cloud.r-project.org/bin/linux/ubuntu bionic-cran40/ InRelease [3,626 B]
Ign:2 http://developer.download.nvidia.com/compute/machine-learning/repos/ubuntu1804/x86_64  InRelease
Hit:3 http://developer.download.nvidia.com/compute/machine-learning/repos/ubuntu1804/x86_64  Release
Ign:4 https://developer.download.nvidia.com/compute/cuda/repos/ubuntu1804/x86_64  InRelease
Get:5 https://developer.download.nvidia.com/compute/cuda/repos/ubuntu1804/x86_64  Release [696 B]
Get:6 https://developer.download.nvidia.com/compute/cuda/repos/ubuntu1804/x86_64  Release.gpg [836 B]
Get:7 https://cloud.r-project.org/bin/linux/ubuntu bionic-cran40/ Packages [82.3 kB]
Hit:8 http://archive.ubuntu.com/ubuntu bionic InRelease
Get:10 http://security.ubuntu.com/ubuntu bionic-security InRelease [88.7 kB]
Get:11 http://ppa.launchpad.net/c2d4u.team/c2d4u4.0+/ubuntu bionic InRelease [15.9 kB]
Get:12 http://archive.ubuntu.com/ubuntu bionic-updates InRelease [88.7 kB]
Get:13 https://developer.download.nvidia.com/compute/cuda/repos/ubuntu1804/x86_64  Packages [952 kB]
Get:14 http://archive.ubuntu.com/ubuntu bionic-backports InRelease [74.6 kB]
```

```
Hit:15 http://ppa.launchpad.net/cran/libgit2/ubuntu bionic InRelease
Get:16 http://security.ubuntu.com/ubuntu bionic-security/universe
amd64 Packages [1,490 kB]
Get:17 http://archive.ubuntu.com/ubuntu bionic-updates/restricted
amd64 Packages [918 kB]
Get:18 http://ppa.launchpad.net/deadsnakes/ppa/ubuntu bionic InRelease
[15.9 kB]
Get:19 http://archive.ubuntu.com/ubuntu bionic-updates/universe amd64
Packages [2,268 kB]
Get:20 http://ppa.launchpad.net/graphics-drivers/ppa/ubuntu bionic
InRelease [21.3 kB]
Get:21 http://security.ubuntu.com/ubuntu bionic-security/restricted
amd64 Packages [884 kB]
Get:22 http://archive.ubuntu.com/ubuntu bionic-updates/multiverse
amd64 Packages [29.8 kB]
Get:23 http://archive.ubuntu.com/ubuntu bionic-updates/main amd64
Packages [3,133 kB]
Get:24 http://security.ubuntu.com/ubuntu bionic-security/main amd64
Packages [2,694 kB]
Get:25 http://archive.ubuntu.com/ubuntu bionic-backports/main amd64
Packages [12.2 kB]
Get:26 http://archive.ubuntu.com/ubuntu bionic-backports/universe
amd64 Packages [12.9 kB]
Get:27 http://ppa.launchpad.net/c2d4u.team/c2d4u4.0+/ubuntu
bionic/main Sources [1,831 kB]
Get:28 http://ppa.launchpad.net/c2d4u.team/c2d4u4.0+/ubuntu
bionic/main amd64 Packages [938 kB]
Get:29 http://ppa.launchpad.net/deadsnakes/ppa/ubuntu bionic/main
amd64 Packages [45.2 kB]
Get:30 http://ppa.launchpad.net/graphics-drivers/ppa/ubuntu
bionic/main amd64 Packages [44.3 kB]
Fetched 15.6 MB in 9s (1,720 kB/s)
Reading package lists...
Reading package lists...
Building dependency tree...
Reading state information...
The following NEW packages will be installed:
  libnvinfer5
0 upgraded, 1 newly installed, 0 to remove and 96 not upgraded.
Need to get 46.7 MB of archives.
After this operation, 163 MB of additional disk space will be used.
Get:1
http://developer.download.nvidia.com/compute/machine-learning/repos/
ubuntu1804/x86_64  libnvinfer5 5.1.5-1+cuda10.1 [46.7 MB]
Fetched 46.7 MB in 3s (13.4 MB/s)
Selecting previously unselected package libnvinfer5.
(Reading database ...  (Reading database ... 5% (Reading database ...
10% (Reading database ... 15% (Reading database ... 20% (Reading
database ... 25% (Reading database ... 30% (Reading database ... 35%
(Reading database ... 40% (Reading database ... 45% (Reading
```

```
database ... 50% (Reading database ... 55% (Reading database ... 60%
(Reading database ... 65% (Reading database ... 70% (Reading
database ... 75% (Reading database ... 80% (Reading database ... 85%
(Reading database ... 90% (Reading database ... 95% (Reading
database ... 100% (Reading database ... 156213 files and directories
currently installed.)
Preparing to unpack .../libnvinfer5_5.1.5-1+cuda10.1_amd64.deb ...
Unpacking libnvinfer5 (5.1.5-1+cuda10.1) ...
Setting up libnvinfer5 (5.1.5-1+cuda10.1) ...
Processing triggers for libc-bin (2.27-3ubuntu1.3) ...
/sbin/ldconfig.real:
/usr/local/lib/python3.7/dist-packages/ideep4py/lib/libmkldnn.so.0 is
not a symbolic link


--2022-04-08 16:26:04--
https://developer.download.nvidia.com/compute/machine-learning/repos/
ubuntu1804/x86_64/nvidia-machine-learning-repo-ubuntu1804_1.0.0-
1_amd64.deb
Resolving developer.download.nvidia.com
(developer.download.nvidia.com)... 152.199.39.144
Connecting to developer.download.nvidia.com
(developer.download.nvidia.com)|152.199.39.144|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 2926 (2.9K) [application/x-deb]
Saving to: ‘nvidia-machine-learning-repo-ubuntu1804_1.0.0-1_amd64.deb’

     0K ..                                                    100%
148M=0s

2022-04-08 16:26:04 (148 MB/s) - ‘nvidia-machine-learning-repo-
ubuntu1804_1.0.0-1_amd64.deb’ saved [2926/2926]

W: Target Packages (Packages) is configured multiple times in
/etc/apt/sources.list.d/nvidia-machine-learning.list:1 and
/etc/apt/sources.list.d/nvidia-ml.list:1
W: Target Packages (Packages) is configured multiple times in
/etc/apt/sources.list.d/nvidia-machine-learning.list:1 and
/etc/apt/sources.list.d/nvidia-ml.list:1
debconf: unable to initialize frontend: Dialog
debconf: (No usable dialog-like program is installed, so the dialog
based frontend cannot be used. at
/usr/share/perl5/Debconf/FrontEnd/Dialog.pm line 76, <> line 1.)
debconf: falling back to frontend: Readline
debconf: unable to initialize frontend: Readline
debconf: (This frontend requires a controlling tty.)
debconf: falling back to frontend: Teletype
dpkg-preconfigure: unable to re-open stdin:
W: Target Packages (Packages) is configured multiple times in
```

```
/etc/apt/sources.list.d/nvidia-machine-learning.list:1 and
/etc/apt/sources.list.d/nvidia-ml.list:1

# check TensorRT version
print("TensorRT version: ")
!dpkg -l | grep nvinfer

TensorRT version:
ii  libnvinfer5                                5.1.5-1+cuda10.1
amd64        TensorRT runtime libraries
```

A successfull TensorRT installation looks like:

```
TensorRT version:
ii  libnvinfer5   5.1.5-1+cuda10.1   amd64        TensorRT runtime
libraries
```

Check Tensor core GPU

The below code check whether a Tensor-core GPU is present.

```
from tensorflow.python.client import device_lib

def check_tensor_core_gpu_present():
    local_device_protos = device_lib.list_local_devices()
    for line in local_device_protos:
        if "compute capability" in str(line):
            compute_capability =
float(line.physical_device_desc.split("compute capability: ")[-1])
            if compute_capability>=7.0:
                return True

print("Tensor Core GPU Present:", check_tensor_core_gpu_present())
tensor_core_gpu = check_tensor_core_gpu_present()

Tensor Core GPU Present: True
```

Importing required libraries

```
from __future__ import absolute_import, division, print_function,
unicode_literals
import os
import time

import numpy as np
import matplotlib.pyplot as plt

import tensorflow as tf
from tensorflow import keras
from tensorflow.python.compiler.tensorrt import trt_convert as trt
from tensorflow.python.saved_model import tag_constants
from tensorflow.keras.applications.resnet50 import ResNet50
from tensorflow.keras.preprocessing import image
```

```
from tensorflow.keras.applications.resnet50 import preprocess_input,
decode_predictions
```

Data

We download several random images for testing from the Internet.

```
image_dir = '/content/drive/MyDrive/Colab Notebooks/train/Black-grass'

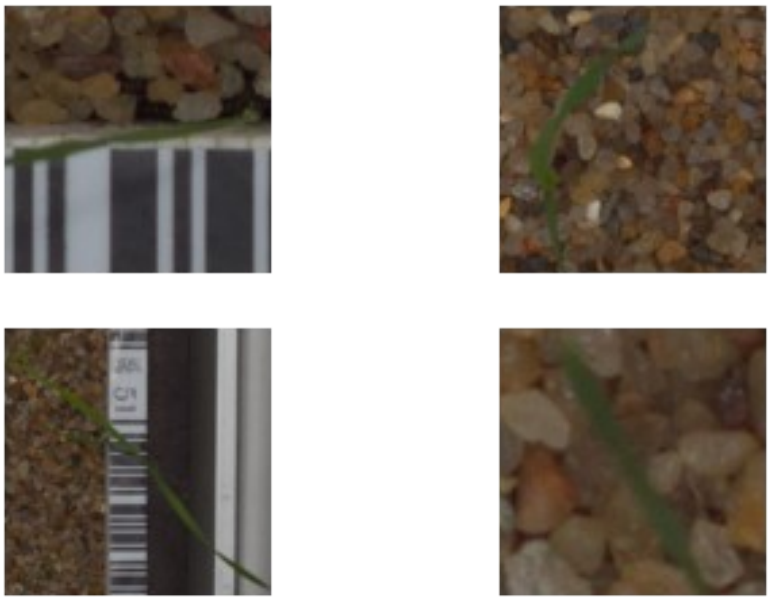
import os
images = os.listdir(image_dir)
images[1]
```

```
{"type":"string"}
```

```
from tensorflow.keras.preprocessing import image

fig, axes = plt.subplots(nrows=2, ncols=2)

for i in range(4):
  img_file = images[i]
  img_path = os.path.join(image_dir,img_file)
  img = image.load_img(img_path, target_size=(224, 224))
  plt.subplot(2,2,i+1)
  plt.imshow(img);
  plt.axis('off');
```



Model

We next download and test a ResNet-50 pre-trained model from the Keras model zoo.

```
model = ResNet50(weights='imagenet')
```

```
Downloading data from https://github.com/keras-team/keras-
applications/releases/download/resnet/
resnet50_weights_tf_dim_ordering_tf_kernels.h5
102973440/102967424 [==============================] - 9s 0us/step
```

```
for i in range(4):
  img_file = images[i]
  img_path = os.path.join(image_dir,img_file)
  img = image.load_img(img_path, target_size=(224, 224))
  x = image.img_to_array(img)
  x = np.expand_dims(x, axis=0)
  x = preprocess_input(x)

  preds = model.predict(x)
  # decode the results into a list of tuples (class, description,
probability)
  # (one such list for each sample in the batch)
  print('{} - Predicted: {}'.format(img_path,
decode_predictions(preds, top=3)[0]))

  plt.subplot(2,2,i+1)
  plt.imshow(img);
  plt.axis('off');
  plt.title(decode_predictions(preds, top=3)[0][0][1])
```

```
Downloading data from
https://storage.googleapis.com/download.tensorflow.org/data/imagenet_c
lass_index.json
40960/35363 [==================================] - 0s 0us/step
/content/drive/MyDrive/Colab Notebooks/train/Black-grass/0050f38b3.png
- Predicted: [('n07880968', 'burrito', 0.3077643), ('n03930313',
'picket_fence', 0.13480566), ('n03089624', 'confectionery',
0.06849264)]
/content/drive/MyDrive/Colab Notebooks/train/Black-grass/0183fdf68.png
- Predicted: [('n01632777', 'axolotl', 0.083475195), ('n01737021',
'water_snake', 0.07512069), ('n01667778', 'terrapin', 0.05346586)]
/content/drive/MyDrive/Colab Notebooks/train/Black-grass/0260cffa8.png
- Predicted: [('n02231487', 'walking_stick', 0.15622911),
('n03871628', 'packet', 0.15378147), ('n02236044', 'mantis',
0.12198411)]
/content/drive/MyDrive/Colab Notebooks/train/Black-grass/05eedce4d.png
```

```
- Predicted: [('n07590611', 'hot_pot', 0.20829123), ('n07875152',
'potpie', 0.19551334), ('n07579787', 'plate', 0.13544671)]
```



TF-TRT takes input as aTensorFlow saved model, therefore, we re-export the Keras model as a TF saved model.

```
# Save the entire model as a SavedModel.
model.save('/content/drive/MyDrive/Colab
Notebooks/resnet50_saved_model')
```

```
WARNING:tensorflow:From
/usr/local/lib/python3.7/dist-packages/tensorflow_core/python/ops/
resource_variable_ops.py:1781: calling BaseResourceVariable.__init__
(from tensorflow.python.ops.resource_variable_ops) with constraint is
deprecated and will be removed in a future version.
Instructions for updating:
If using Keras pass *_constraint arguments to layers.
INFO:tensorflow:Assets written to: /content/drive/MyDrive/Colab
Notebooks/resnet50_saved_model/assets
```

```
!saved_model_cli show --all --dir resnet50_saved_model
```

```
Traceback (most recent call last):
  File "/usr/local/bin/saved_model_cli", line 8, in <module>
    sys.exit(main())
  File
"/usr/local/lib/python3.7/dist-packages/tensorflow_core/python/tools/
saved_model_cli.py", line 909, in main
    args.func(args)
  File
"/usr/local/lib/python3.7/dist-packages/tensorflow_core/python/tools/
```

```
saved_model_cli.py", line 610, in show
    _show_all(args.dir)
  File
"/usr/local/lib/python3.7/dist-packages/tensorflow_core/python/tools/
saved_model_cli.py", line 192, in _show_all
    tag_sets =
saved_model_utils.get_saved_model_tag_sets(saved_model_dir)
  File
"/usr/local/lib/python3.7/dist-packages/tensorflow_core/python/tools/
saved_model_utils.py", line 88, in get_saved_model_tag_sets
    saved_model = read_saved_model(saved_model_dir)
  File
"/usr/local/lib/python3.7/dist-packages/tensorflow_core/python/tools/
saved_model_utils.py", line 55, in read_saved_model
    raise IOError("SavedModel file does not exist at: %s" %
saved_model_dir)
OSError: SavedModel file does not exist at: resnet50_saved_model
```

Inference with native TF2.0 saved model

```
model = tf.keras.models.load_model('/content/drive/MyDrive/Colab
Notebooks/resnet50_saved_model')

img_file = images[i]
img_path = os.path.join(image_dir,img_file)
img = image.load_img(img_path, target_size=(224, 224))
x = image.img_to_array(img)
x = np.expand_dims(x, axis=0)
x = preprocess_input(x)

preds = model.predict(x)
# decode the results into a list of tuples (class, description,
probability)
# (one such list for each sample in the batch)
print('{} - Predicted: {}'.format(img_path, decode_predictions(preds,
top=3)[0]))
plt.subplot(2,2,1)
plt.imshow(img);
plt.axis('off');
plt.title(decode_predictions(preds, top=3)[0][0][1])
```

```
/content/drive/MyDrive/Colab Notebooks/train/Black-grass/05eedce4d.png
- Predicted: [('n07590611', 'hot_pot', 0.2082909), ('n07875152',
'potpie', 0.1955132), ('n07579787', 'plate', 0.13544662)]
```

```
Text(0.5, 1.0, 'hot_pot')
```

hot_pot



```
batch_size = 8
batched_input = np.zeros((batch_size, 224, 224, 3), dtype=np.float32)

for i in range(batch_size):
  img_file = images[i%4]
  img_path = os.path.join(image_dir,img_file)
  img = image.load_img(img_path, target_size=(224, 224))
  x = image.img_to_array(img)
  x = np.expand_dims(x, axis=0)
  x = preprocess_input(x)
  batched_input[i, :] = x
batched_input = tf.constant(batched_input)
print('batched_input shape: ', batched_input.shape)
```

```
batched_input shape:  (8, 224, 224, 3)
```

```
# Benchmarking throughput

N_run = 300
elapsed_time = []

for i in range(N_run):
  start_time = time.time()
  preds = model.predict(batched_input)
  end_time = time.time()
  elapsed_time = np.append(elapsed_time, end_time - start_time)
  if i % 50 == 0:
    print('Step {}: {:4.1f}ms'.format(i, (elapsed_time[-50:].mean()) *
1000))

print('Throughput: {:.0f} images/s'.format(N_run * batch_size /
elapsed_time.sum()))
```

```
Step 0: 5427.2ms
Step 50: 86.0ms
Step 100: 85.5ms
Step 150: 85.4ms
Step 200: 86.4ms
```

```
Step 250: 86.2ms
Throughput: 77 images/s
```

TF-TRT FP32 model

We first convert the TF native FP32 model to a TF-TRT FP32 model.

```
print('Converting to TF-TRT FP32...')
conversion_params =
trt.DEFAULT_TRT_CONVERSION_PARAMS._replace(precision_mode=trt.TrtPreci
sionMode.FP32,

max_workspace_size_bytes=8000000000)

converter =
trt.TrtGraphConverterV2(input_saved_model_dir='resnet50_saved_model',

conversion_params=conversion_params)
converter.convert()
converter.save(output_saved_model_dir='resnet50_saved_model_TFTRT_FP32
')
print('Done Converting to TF-TRT FP32')
```

```
Converting to TF-TRT FP32...
INFO:tensorflow:Linked TensorRT version: (5, 1, 5)
INFO:tensorflow:Loaded TensorRT version: (5, 1, 5)
INFO:tensorflow:Running against TensorRT version 5.1.5
INFO:tensorflow:Assets written to:
resnet50_saved_model_TFTRT_FP32/assets
Done Converting to TF-TRT FP32
```

```
!saved_model_cli show --all --dir resnet50_saved_model_TFTRT_FP32
```

```
MetaGraphDef with tag-set: 'serve' contains the following
SignatureDefs:

signature_def['__saved_model_init_op']:
  The given SavedModel SignatureDef contains the following input(s):
  The given SavedModel SignatureDef contains the following output(s):
    outputs['__saved_model_init_op'] tensor_info:
        dtype: DT_INVALID
        shape: unknown_rank
        name: NoOp
  Method name is:

signature_def['serving_default']:
  The given SavedModel SignatureDef contains the following input(s):
    inputs['input_1'] tensor_info:
        dtype: DT_FLOAT
        shape: (-1, 224, 224, 3)
```

```
        name: serving_default_input_1:0
  The given SavedModel SignatureDef contains the following output(s):
    outputs['probs'] tensor_info:
        dtype: DT_FLOAT
        shape: unknown_rank
        name: PartitionedCall:0
  Method name is: tensorflow/serving/predict
```

Next, we load and test the TF-TRT FP32 model.

```
def predict_tftrt(input_saved_model):
    """Runs prediction on a single image and shows the result.
    input_saved_model (string): Name of the input model stored in the
current dir
    """
    img_file = images[0]
    img_path = os.path.join(image_dir,img_file)
    img = image.load_img(img_path, target_size=(224, 224))
    x = image.img_to_array(img)
    x = np.expand_dims(x, axis=0)
    x = preprocess_input(x)
    x = tf.constant(x)

    saved_model_loaded = tf.saved_model.load(input_saved_model,
tags=[tag_constants.SERVING])
    signature_keys = list(saved_model_loaded.signatures.keys())
    print(signature_keys)

    infer = saved_model_loaded.signatures['serving_default']
    print(infer.structured_outputs)

    labeling = infer(x)
    preds = labeling['probs'].numpy()
    print('{} - Predicted: {}'.format(img_path,
decode_predictions(preds, top=3)[0]))
    plt.subplot(2,2,1)
    plt.imshow(img);
    plt.axis('off');
    plt.title(decode_predictions(preds, top=3)[0][0][1])

predict_tftrt('resnet50_saved_model_TFTRT_FP32')
```

```
['serving_default']
{'probs': TensorSpec(shape=<unknown>, dtype=tf.float32, name='probs')}
/content/drive/MyDrive/Colab Notebooks/train/Black-grass/0050f38b3.png
- Predicted: [('n07880968', 'burrito', 0.30776346), ('n03930313',
'picket_fence', 0.1348056), ('n03089624', 'confectionery',
0.06849259)]
```




```
def benchmark_tftrt(input_saved_model):
    saved_model_loaded = tf.saved_model.load(input_saved_model,
tags=[tag_constants.SERVING])
    infer = saved_model_loaded.signatures['serving_default']


    N_run = 300
    elapsed_time = []

    for i in range(N_run):
      start_time = time.time()
      labeling = infer(batched_input)
      #prob = labeling['probs'].numpy()
      end_time = time.time()
      elapsed_time = np.append(elapsed_time, end_time - start_time)
      if i % 50 == 0:
        print('Step {}: {:4.1f}ms'.format(i, (elapsed_time[-
50:].mean()) * 1000))

    print('Throughput: {:.0f} images/s'.format(N_run * batch_size /
elapsed_time.sum()))

benchmark_tftrt('resnet50_saved_model_TFTRT_FP32')
```

```
Step 0: 8794.7ms
Step 50: 61.8ms
Step 100: 62.1ms
Step 150: 62.1ms
Step 200: 62.3ms
Step 250: 62.0ms
Throughput: 88 images/s
```

TF-TRT FP16 model

We next convert the native TF FP32 saved model to TF-TRT FP16 model.

```
print('Converting to TF-TRT FP16...')
conversion_params = trt.DEFAULT_TRT_CONVERSION_PARAMS._replace(
    precision_mode=trt.TrtPrecisionMode.FP16,
    max_workspace_size_bytes=8000000000)
```

```
converter = trt.TrtGraphConverterV2(
   input_saved_model_dir='/content/drive/MyDrive/Colab
Notebooks/resnet50_saved_model', conversion_params=conversion_params)
converter.convert()
converter.save(output_saved_model_dir='resnet50_saved_model_TFTRT_FP16
')
print('Done Converting to TF-TRT FP16')
```

```
Converting to TF-TRT FP16...
INFO:tensorflow:Linked TensorRT version: (5, 1, 5)
INFO:tensorflow:Loaded TensorRT version: (5, 1, 5)
INFO:tensorflow:Running against TensorRT version 5.1.5
INFO:tensorflow:Assets written to:
resnet50_saved_model_TFTRT_FP16/assets
Done Converting to TF-TRT FP16
```

```
predict_tftrt('resnet50_saved_model_TFTRT_FP16')
```

```
['serving_default']
{'probs': TensorSpec(shape=<unknown>, dtype=tf.float32, name='probs')}
/content/drive/MyDrive/Colab Notebooks/train/Black-grass/0050f38b3.png
- Predicted: [('n07880968', 'burrito', 0.30785987), ('n03930313',
'picket_fence', 0.13485312), ('n03089624', 'confectionery',
0.06875941)]
```




```
benchmark_tftrt('resnet50_saved_model_TFTRT_FP16')
```

```
Step 0: 25238.4ms
Step 50: 52.6ms
Step 100: 52.8ms
Step 150: 53.3ms
Step 200: 53.1ms
Step 250: 52.9ms
Throughput: 58 images/s
```

TF-TRT INT8 model

Creating TF-TRT INT8 model requires a small calibration dataset. This data set ideally should represent the test data in production well, and will be used to create a value histogram for each layer in the neural network for effective 8-bit quantization.

Herein, for demonstration purposes, we take only the 4 images that we downloaded for calibration. In production, this set should be more representative of the production data.

Due to Colab memory limit which sometime causes TensorRT to crash, to proceed, first, restart the runtime by pressing CTRL+M or select *Runtime -> Restart runtime...* or simply execute the next cell.

```
import os
os.kill(os.getpid(), 9)
```

```
from __future__ import absolute_import, division, print_function,
unicode_literals
import os
import time

import numpy as np
import matplotlib.pyplot as plt

import tensorflow as tf
from tensorflow import keras
from tensorflow.python.compiler.tensorrt import trt_convert as trt
from tensorflow.python.saved_model import tag_constants
from tensorflow.keras.applications.resnet50 import ResNet50
from tensorflow.keras.preprocessing import image
from tensorflow.keras.applications.resnet50 import preprocess_input,
decode_predictions
```

```
batch_size = 8
batched_input = np.zeros((batch_size, 224, 224, 3), dtype=np.float32)

for i in range(batch_size):
  img_file = images[i%4]
  img_path = os.path.join(image_dir,img_file)
  img = image.load_img(img_path, target_size=(224, 224))
  x = image.img_to_array(img)
  x = np.expand_dims(x, axis=0)
  x = preprocess_input(x)
  batched_input[i, :] = x
batched_input = tf.constant(batched_input)
print('batched_input shape: ', batched_input.shape)
```

```
batched_input shape:  (8, 224, 224, 3)
```

```
print('Converting to TF-TRT INT8...')
conversion_params = trt.DEFAULT_TRT_CONVERSION_PARAMS._replace(
    precision_mode=trt.TrtPrecisionMode.INT8,
    max_workspace_size_bytes=8000000000,
    use_calibration=True)
```

```
converter = trt.TrtGraphConverterV2(
    input_saved_model_dir='/content/drive/MyDrive/Colab
Notebooks/resnet50_saved_model',
    conversion_params=conversion_params)

def calibration_input_fn():
    yield (batched_input, )
converter.convert(calibration_input_fn=calibration_input_fn)

converter.save(output_saved_model_dir='resnet50_saved_model_TFTRT_INT8
')
print('Done Converting to TF-TRT INT8')
```

```
Converting to TF-TRT INT8...
INFO:tensorflow:Linked TensorRT version: (5, 1, 5)
INFO:tensorflow:Loaded TensorRT version: (5, 1, 5)
INFO:tensorflow:Running against TensorRT version 5.1.5
WARNING:tensorflow:From
/usr/local/lib/python3.7/dist-packages/tensorflow_core/python/ops/
resource_variable_ops.py:1781: calling BaseResourceVariable.__init__
(from tensorflow.python.ops.resource_variable_ops) with constraint is
deprecated and will be removed in a future version.
Instructions for updating:
If using Keras pass *_constraint arguments to layers.
INFO:tensorflow:Assets written to:
resnet50_saved_model_TFTRT_INT8/assets
Done Converting to TF-TRT INT8
```

```
def predict_tftrt(input_saved_model):
    """Runs prediction on a single image and shows the result.
    input_saved_model (string): Name of the input model stored in the
current dir
    """
    img_file = images[0]
    img_path = os.path.join(image_dir,img_file)
    img = image.load_img(img_path, target_size=(224, 224))
    x = image.img_to_array(img)
    x = np.expand_dims(x, axis=0)
    x = preprocess_input(x)
    x = tf.constant(x)

    saved_model_loaded = tf.saved_model.load(input_saved_model,
tags=[tag_constants.SERVING])
    signature_keys = list(saved_model_loaded.signatures.keys())
    print(signature_keys)

    infer = saved_model_loaded.signatures['serving_default']
    print(infer.structured_outputs)

    labeling = infer(x)
```

```
    preds = labeling['probs'].numpy()
    print('{} - Predicted: {}'.format(img_path,
decode_predictions(preds, top=3)[0]))
    plt.subplot(2,2,1)
    plt.imshow(img);
    plt.axis('off');
    plt.title(decode_predictions(preds, top=3)[0][0][1])

predict_tftrt('resnet50_saved_model_TFTRT_INT8')
```

```
['serving_default']
{'probs': TensorSpec(shape=<unknown>, dtype=tf.float32, name='probs')}
/content/drive/MyDrive/Colab Notebooks/train/Black-grass/0050f38b3.png
- Predicted: [('n03930313', 'picket_fence', 0.21861497), ('n03089624',
'confectionery', 0.21403234), ('n07880968', 'burrito', 0.098421164)]
```




```
def benchmark_tftrt(input_saved_model):
    saved_model_loaded = tf.saved_model.load(input_saved_model,
tags=[tag_constants.SERVING])
    infer = saved_model_loaded.signatures['serving_default']

    N_run = 300
    elapsed_time = []

    for i in range(N_run):
      start_time = time.time()
      labeling = infer(batched_input)
      #prob = labeling['probs'].numpy()
      end_time = time.time()
      elapsed_time = np.append(elapsed_time, end_time - start_time)
      if i % 50 == 0:
        print('Step {}: {:4.1f}ms'.format(i, (elapsed_time[-
50:].mean()) * 1000))

    print('Throughput: {:.0f} images/s'.format(N_run * batch_size /
elapsed_time.sum()))

benchmark_tftrt('resnet50_saved_model_TFTRT_INT8')
```

```
WARNING:tensorflow:Unresolved object in checkpoint:
(root).trt_engine_resources.TRTEngineOp_0._serialized_trt_resource_fil
ename
WARNING:tensorflow:A checkpoint was restored (e.g.
tf.train.Checkpoint.restore or tf.keras.Model.load_weights) but not
all checkpointed values were used. See above for specific issues. Use
expect_partial() on the load status object, e.g.
tf.train.Checkpoint.restore(...).expect_partial(), to silence these
warnings, or use assert_consumed() to make the check explicit. See
https://www.tensorflow.org/alpha/guide/checkpoints#loading_mechanics
for details.
Step 0: 1442.1ms
Step 50:  5.8ms
Step 100:  3.4ms
Step 150:  3.4ms
Step 200:  3.3ms
Step 250:  3.3ms
Throughput: 934 images/s
```

Conclusion

In this notebook, we have demonstrated the process of creating TF-TRT FP32, FP16 and INT8 inference models from an original Keras FP32 model, as well as verify their speed and accuracy.

What's next

Try TF-TRT on your own model and data, and experience the simplicity and speed up it offers.

```
!pip install pillow matplotlib
!pip install tensorflow-gpu==2.0.0
```

```
Requirement already satisfied: pillow in
/usr/local/lib/python3.7/dist-packages (7.1.2)
Requirement already satisfied: matplotlib in
/usr/local/lib/python3.7/dist-packages (3.2.2)
Requirement already satisfied: numpy>=1.11 in
/usr/local/lib/python3.7/dist-packages (from matplotlib) (1.21.5)
Requirement already satisfied: kiwisolver>=1.0.1 in
/usr/local/lib/python3.7/dist-packages (from matplotlib) (1.4.0)
Requirement already satisfied: python-dateutil>=2.1 in
/usr/local/lib/python3.7/dist-packages (from matplotlib) (2.8.2)
Requirement already satisfied: cycler>=0.10 in
/usr/local/lib/python3.7/dist-packages (from matplotlib) (0.11.0)
Requirement already satisfied: pyparsing!=2.0.4,!=2.1.2,!
=2.1.6,>=2.0.1 in /usr/local/lib/python3.7/dist-packages (from
matplotlib) (3.0.7)
Requirement already satisfied: typing-extensions in
/usr/local/lib/python3.7/dist-packages (from kiwisolver>=1.0.1-
>matplotlib) (3.10.0.2)
Requirement already satisfied: six>=1.5 in
/usr/local/lib/python3.7/dist-packages (from python-dateutil>=2.1-
>matplotlib) (1.15.0)
Collecting tensorflow-gpu==2.0.0
  Downloading tensorflow_gpu-2.0.0-cp37-cp37m-manylinux2010_x86_64.whl
(380.8 MB)
ent already satisfied: google-pasta>=0.1.6 in
/usr/local/lib/python3.7/dist-packages (from tensorflow-gpu==2.0.0)
(0.2.0)
Collecting tensorboard<2.1.0,>=2.0.0
  Downloading tensorboard-2.0.2-py3-none-any.whl (3.8 MB)
ent already satisfied: numpy<2.0,>=1.16.0 in
/usr/local/lib/python3.7/dist-packages (from tensorflow-gpu==2.0.0)
(1.21.5)
Collecting tensorflow-estimator<2.1.0,>=2.0.0
  Downloading tensorflow_estimator-2.0.1-py2.py3-none-any.whl (449 kB)
ent already satisfied: termcolor>=1.1.0 in
/usr/local/lib/python3.7/dist-packages (from tensorflow-gpu==2.0.0)
(1.1.0)
Collecting gast==0.2.2
  Downloading gast-0.2.2.tar.gz (10 kB)
Requirement already satisfied: opt-einsum>=2.3.2 in
/usr/local/lib/python3.7/dist-packages (from tensorflow-gpu==2.0.0)
(3.3.0)
Requirement already satisfied: keras-preprocessing>=1.0.5 in
/usr/local/lib/python3.7/dist-packages (from tensorflow-gpu==2.0.0)
(1.1.2)
Requirement already satisfied: grpcio>=1.8.6 in
/usr/local/lib/python3.7/dist-packages (from tensorflow-gpu==2.0.0)
```

```
(1.44.0)
Requirement already satisfied: absl-py>=0.7.0 in
/usr/local/lib/python3.7/dist-packages (from tensorflow-gpu==2.0.0)
(1.0.0)
Requirement already satisfied: protobuf>=3.6.1 in
/usr/local/lib/python3.7/dist-packages (from tensorflow-gpu==2.0.0)
(3.17.3)
Requirement already satisfied: wheel>=0.26 in
/usr/local/lib/python3.7/dist-packages (from tensorflow-gpu==2.0.0)
(0.37.1)
Requirement already satisfied: six>=1.10.0 in
/usr/local/lib/python3.7/dist-packages (from tensorflow-gpu==2.0.0)
(1.15.0)
Requirement already satisfied: wrapt>=1.11.1 in
/usr/local/lib/python3.7/dist-packages (from tensorflow-gpu==2.0.0)
(1.14.0)
Requirement already satisfied: astor>=0.6.0 in
/usr/local/lib/python3.7/dist-packages (from tensorflow-gpu==2.0.0)
(0.8.1)
Collecting keras-applications>=1.0.8
  Downloading Keras_Applications-1.0.8-py3-none-any.whl (50 kB)
ent already satisfied: h5py in /usr/local/lib/python3.7/dist-packages
(from keras-applications>=1.0.8->tensorflow-gpu==2.0.0) (3.1.0)
Requirement already satisfied: requests<3,>=2.21.0 in
/usr/local/lib/python3.7/dist-packages (from
tensorboard<2.1.0,>=2.0.0->tensorflow-gpu==2.0.0) (2.23.0)
Requirement already satisfied: google-auth-oauthlib<0.5,>=0.4.1 in
/usr/local/lib/python3.7/dist-packages (from
tensorboard<2.1.0,>=2.0.0->tensorflow-gpu==2.0.0) (0.4.6)
Requirement already satisfied: werkzeug>=0.11.15 in
/usr/local/lib/python3.7/dist-packages (from
tensorboard<2.1.0,>=2.0.0->tensorflow-gpu==2.0.0) (1.0.1)
Requirement already satisfied: setuptools>=41.0.0 in
/usr/local/lib/python3.7/dist-packages (from
tensorboard<2.1.0,>=2.0.0->tensorflow-gpu==2.0.0) (57.4.0)
Requirement already satisfied: markdown>=2.6.8 in
/usr/local/lib/python3.7/dist-packages (from
tensorboard<2.1.0,>=2.0.0->tensorflow-gpu==2.0.0) (3.3.6)
Requirement already satisfied: google-auth<2,>=1.6.3 in
/usr/local/lib/python3.7/dist-packages (from
tensorboard<2.1.0,>=2.0.0->tensorflow-gpu==2.0.0) (1.35.0)
Requirement already satisfied: rsa<5,>=3.1.4 in
/usr/local/lib/python3.7/dist-packages (from google-auth<2,>=1.6.3-
>tensorboard<2.1.0,>=2.0.0->tensorflow-gpu==2.0.0) (4.8)
Requirement already satisfied: pyasn1-modules>=0.2.1 in
/usr/local/lib/python3.7/dist-packages (from google-auth<2,>=1.6.3-
>tensorboard<2.1.0,>=2.0.0->tensorflow-gpu==2.0.0) (0.2.8)
Requirement already satisfied: cachetools<5.0,>=2.0.0 in
/usr/local/lib/python3.7/dist-packages (from google-auth<2,>=1.6.3-
>tensorboard<2.1.0,>=2.0.0->tensorflow-gpu==2.0.0) (4.2.4)
```

```
Requirement already satisfied: requests-oauthlib>=0.7.0 in
/usr/local/lib/python3.7/dist-packages (from google-auth-
oauthlib<0.5,>=0.4.1->tensorboard<2.1.0,>=2.0.0->tensorflow-
gpu==2.0.0) (1.3.1)
Requirement already satisfied: importlib-metadata>=4.4 in
/usr/local/lib/python3.7/dist-packages (from markdown>=2.6.8-
>tensorboard<2.1.0,>=2.0.0->tensorflow-gpu==2.0.0) (4.11.3)
Requirement already satisfied: zipp>=0.5 in
/usr/local/lib/python3.7/dist-packages (from importlib-metadata>=4.4-
>markdown>=2.6.8->tensorboard<2.1.0,>=2.0.0->tensorflow-gpu==2.0.0)
(3.7.0)
Requirement already satisfied: typing-extensions>=3.6.4 in
/usr/local/lib/python3.7/dist-packages (from importlib-metadata>=4.4-
>markdown>=2.6.8->tensorboard<2.1.0,>=2.0.0->tensorflow-gpu==2.0.0)
(3.10.0.2)
Requirement already satisfied: pyasn1<0.5.0,>=0.4.6 in
/usr/local/lib/python3.7/dist-packages (from pyasn1-modules>=0.2.1-
>google-auth<2,>=1.6.3->tensorboard<2.1.0,>=2.0.0->tensorflow-
gpu==2.0.0) (0.4.8)
Requirement already satisfied: urllib3!=1.25.0,!=1.25.1,<1.26,>=1.21.1
in /usr/local/lib/python3.7/dist-packages (from requests<3,>=2.21.0-
>tensorboard<2.1.0,>=2.0.0->tensorflow-gpu==2.0.0) (1.24.3)
Requirement already satisfied: idna<3,>=2.5 in
/usr/local/lib/python3.7/dist-packages (from requests<3,>=2.21.0-
>tensorboard<2.1.0,>=2.0.0->tensorflow-gpu==2.0.0) (2.10)
Requirement already satisfied: certifi>=2017.4.17 in
/usr/local/lib/python3.7/dist-packages (from requests<3,>=2.21.0-
>tensorboard<2.1.0,>=2.0.0->tensorflow-gpu==2.0.0) (2021.10.8)
Requirement already satisfied: chardet<4,>=3.0.2 in
/usr/local/lib/python3.7/dist-packages (from requests<3,>=2.21.0-
>tensorboard<2.1.0,>=2.0.0->tensorflow-gpu==2.0.0) (3.0.4)
Requirement already satisfied: oauthlib>=3.0.0 in
/usr/local/lib/python3.7/dist-packages (from requests-oauthlib>=0.7.0-
>google-auth-oauthlib<0.5,>=0.4.1->tensorboard<2.1.0,>=2.0.0-
>tensorflow-gpu==2.0.0) (3.2.0)
Requirement already satisfied: cached-property in
/usr/local/lib/python3.7/dist-packages (from h5py->keras-
applications>=1.0.8->tensorflow-gpu==2.0.0) (1.5.2)
Building wheels for collected packages: gast
  Building wheel for gast (setup.py) ... e=gast-0.2.2-py3-none-any.whl
size=7554
sha256=16719dd00ceaa09f8b078b9b2c64abc5b3b9efafbdb2c8711b8f1c820f5dd79
0
  Stored in directory:
/root/.cache/pip/wheels/21/7f/02/420f32a803f7d0967b48dd823da3f558c5166
991bfd204eef3
Successfully built gast
Installing collected packages: tensorflow-estimator, tensorboard,
keras-applications, gast, tensorflow-gpu
  Attempting uninstall: tensorflow-estimator
```

```
    Found existing installation: tensorflow-estimator 2.8.0
    Uninstalling tensorflow-estimator-2.8.0:
      Successfully uninstalled tensorflow-estimator-2.8.0
  Attempting uninstall: tensorboard
    Found existing installation: tensorboard 2.8.0
    Uninstalling tensorboard-2.8.0:
      Successfully uninstalled tensorboard-2.8.0
  Attempting uninstall: gast
    Found existing installation: gast 0.5.3
    Uninstalling gast-0.5.3:
      Successfully uninstalled gast-0.5.3
ERROR: pip's dependency resolver does not currently take into account
all the packages that are installed. This behaviour is the source of
the following dependency conflicts.
tensorflow 2.8.0 requires tf-estimator-nightly==2.8.0.dev2021122109,
which is not installed.
tensorflow 2.8.0 requires tensorboard<2.9,>=2.8, but you have
tensorboard 2.0.2 which is incompatible.
tensorflow-probability 0.16.0 requires gast>=0.3.2, but you have gast
0.2.2 which is incompatible.
Successfully installed gast-0.2.2 keras-applications-1.0.8
tensorboard-2.0.2 tensorflow-estimator-2.0.1 tensorflow-gpu-2.0.0
```

```
!pip install 'h5py==2.10.0'
```

```
Collecting h5py==2.10.0
  Downloading h5py-2.10.0-cp37-cp37m-manylinux1_x86_64.whl (2.9 MB)
ent already satisfied: six in /usr/local/lib/python3.7/dist-packages
(from h5py==2.10.0) (1.15.0)
Requirement already satisfied: numpy>=1.7 in
/usr/local/lib/python3.7/dist-packages (from h5py==2.10.0) (1.21.5)
Installing collected packages: h5py
  Attempting uninstall: h5py
    Found existing installation: h5py 3.1.0
    Uninstalling h5py-3.1.0:
      Successfully uninstalled h5py-3.1.0
ERROR: pip's dependency resolver does not currently take into account
all the packages that are installed. This behaviour is the source of
the following dependency conflicts.
tensorflow 2.8.0 requires tf-estimator-nightly==2.8.0.dev2021122109,
which is not installed.
tensorflow 2.8.0 requires tensorboard<2.9,>=2.8, but you have
tensorboard 2.0.2 which is incompatible.
Successfully installed h5py-2.10.0
```

```
from google.colab import drive
drive.mount('/content/drive')
```

```
Mounted at /content/drive
```

```
# Copyright 2019 NVIDIA Corporation. All Rights Reserved.
#
# Licensed under the Apache License, Version 2.0 (the "License");
# you may not use this file except in compliance with the License.
# You may obtain a copy of the License at
#
#     http://www.apache.org/licenses/LICENSE-2.0
#
# Unless required by applicable law or agreed to in writing, software
# distributed under the License is distributed on an "AS IS" BASIS,
# WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or
implied.
# See the License for the specific language governing permissions and
# limitations under the License.
#
================================================================================
========
```


TF-TRT Inference from Keras Model with TensorFlow 2.0

Introduction

The NVIDIA TensorRT is a C++ library that facilitates high performance inference on NVIDIA graphics processing units (GPUs). TensorRT takes a trained network, which consists of a network definition and a set of trained parameters, and produces a highly optimized runtime engine which performs inference for that network.

TensorFlow™ integration with TensorRT™ (TF-TRT) optimizes and executes compatible subgraphs, allowing TensorFlow to execute the remaining graph. While you can still use TensorFlow's wide and flexible feature set, TensorRT will parse the model and apply optimizations to the portions of the graph wherever possible.

In this notebook, we demonstrate the process of creating a TF-TRT optimized model from a Inception Keras saved model.

Requirement

GPU

Before running this notebook, please set the Colab runtime environment to GPU via the menu *Runtime => Change runtime type => GPU*.

This demo will work on any NVIDIA GPU with CUDA cores, though for improved FP16 and INT8 inference, a Volta, Turing or newer generation GPU with Tensor cores is desired. On Google Colab, this normally means a T4 GPU. If you are assigned an older K80 GPU, another trial at another time might give you a T4 GPU.

```
!nvidia-smi

Sat Apr  9 07:19:37 2022
+-----------------------------------------------------------------------------
--------+
| NVIDIA-SMI 460.32.03    Driver Version: 460.32.03    CUDA Version:
11.2     |
|-------------------------------+----------------------
+----------------------+
| GPU  Name        Persistence-M| Bus-Id        Disp.A | Volatile
Uncorr. ECC |
| Fan  Temp  Perf  Pwr:Usage/Cap|         Memory-Usage | GPU-Util
Compute M. |
|                               |                      |
MIG M. |
|
===============================+======================+===============
=======|
|   0  Tesla K80           Off  | 00000000:00:04.0 Off |
0 |
| N/A   65C    P8    32W / 149W |      0MiB / 11441MiB |      0%
Default |
|                               |                      |
N/A |
+-------------------------------+----------------------
+----------------------+


+-----------------------------------------------------------------------------
--------+
| Processes:
|
|  GPU   GI   CI        PID   Type   Process name                  GPU
Memory |
|        ID   ID
Usage      |
|
=============================================================================
=======|
|  No running processes found
|
+-----------------------------------------------------------------------------
--------+
```

Install TensorFlow-GPU 2.0

```
import tensorflow as tf
print("Tensorflow version: ", tf.version.VERSION)

Tensorflow version:  2.0.0
```

Install TensorRT Runtime

```
%%bash
wget https://developer.download.nvidia.com/compute/machine-learning/repos/ubuntu1804/x86_64/nvidia-machine-learning-repo-ubuntu1804_1.0.0-1_amd64.deb

dpkg -i nvidia-machine-learning-repo-*.deb
apt-get update

sudo apt-get install libnvinfer5
```

```
Selecting previously unselected package nvidia-machine-learning-repo-ubuntu1804.
(Reading database ... 156210 files and directories currently installed.)
Preparing to unpack nvidia-machine-learning-repo-ubuntu1804_1.0.0-1_amd64.deb ...
Unpacking nvidia-machine-learning-repo-ubuntu1804 (1.0.0-1) ...
Setting up nvidia-machine-learning-repo-ubuntu1804 (1.0.0-1) ...
Get:1 https://cloud.r-project.org/bin/linux/ubuntu bionic-cran40/ InRelease [3,626 B]
Ign:2 http://developer.download.nvidia.com/compute/machine-learning/repos/ubuntu1804/x86_64  InRelease
Hit:3 http://developer.download.nvidia.com/compute/machine-learning/repos/ubuntu1804/x86_64  Release
Ign:4 https://developer.download.nvidia.com/compute/cuda/repos/ubuntu1804/x86_64  InRelease
Get:5 https://developer.download.nvidia.com/compute/cuda/repos/ubuntu1804/x86_64  Release [696 B]
Get:6 https://developer.download.nvidia.com/compute/cuda/repos/ubuntu1804/x86_64  Release.gpg [836 B]
Get:7 https://cloud.r-project.org/bin/linux/ubuntu bionic-cran40/ Packages [82.3 kB]
Get:8 http://security.ubuntu.com/ubuntu bionic-security InRelease [88.7 kB]
Get:10 http://ppa.launchpad.net/c2d4u.team/c2d4u4.0+/ubuntu bionic InRelease [15.9 kB]
Get:11 https://developer.download.nvidia.com/compute/cuda/repos/ubuntu1804/x86_64  Packages [952 kB]
Hit:12 http://archive.ubuntu.com/ubuntu bionic InRelease
Get:13 http://archive.ubuntu.com/ubuntu bionic-updates InRelease [88.7 kB]
Hit:14 http://ppa.launchpad.net/cran/libgit2/ubuntu bionic InRelease
Get:15 http://security.ubuntu.com/ubuntu bionic-security/universe
```

```
amd64 Packages [1,490 kB]
Get:16 http://ppa.launchpad.net/deadsnakes/ppa/ubuntu bionic InRelease
[15.9 kB]
Get:17 http://archive.ubuntu.com/ubuntu bionic-backports InRelease
[74.6 kB]
Get:18 http://archive.ubuntu.com/ubuntu bionic-updates/universe amd64
Packages [2,268 kB]
Get:19 http://ppa.launchpad.net/graphics-drivers/ppa/ubuntu bionic
InRelease [21.3 kB]
Get:20 http://security.ubuntu.com/ubuntu bionic-security/main amd64
Packages [2,694 kB]
Get:21 http://security.ubuntu.com/ubuntu bionic-security/restricted
amd64 Packages [884 kB]
Get:22 http://archive.ubuntu.com/ubuntu bionic-updates/main amd64
Packages [3,133 kB]
Get:23 http://ppa.launchpad.net/c2d4u.team/c2d4u4.0+/ubuntu
bionic/main Sources [1,831 kB]
Get:24 http://archive.ubuntu.com/ubuntu bionic-updates/multiverse
amd64 Packages [29.8 kB]
Get:25 http://archive.ubuntu.com/ubuntu bionic-updates/restricted
amd64 Packages [918 kB]
Get:26 http://archive.ubuntu.com/ubuntu bionic-backports/main amd64
Packages [12.2 kB]
Get:27 http://archive.ubuntu.com/ubuntu bionic-backports/universe
amd64 Packages [12.9 kB]
Get:28 http://ppa.launchpad.net/c2d4u.team/c2d4u4.0+/ubuntu
bionic/main amd64 Packages [938 kB]
Get:29 http://ppa.launchpad.net/deadsnakes/ppa/ubuntu bionic/main
amd64 Packages [45.2 kB]
Get:30 http://ppa.launchpad.net/graphics-drivers/ppa/ubuntu
bionic/main amd64 Packages [44.3 kB]
Fetched 15.6 MB in 9s (1,705 kB/s)
Reading package lists...
Reading package lists...
Building dependency tree...
Reading state information...
The following NEW packages will be installed:
  libnvinfer5
0 upgraded, 1 newly installed, 0 to remove and 96 not upgraded.
Need to get 46.7 MB of archives.
After this operation, 163 MB of additional disk space will be used.
Get:1
http://developer.download.nvidia.com/compute/machine-learning/repos/
ubuntu1804/x86_64  libnvinfer5 5.1.5-1+cuda10.1 [46.7 MB]
Fetched 46.7 MB in 1s (36.1 MB/s)
Selecting previously unselected package libnvinfer5.
(Reading database ...  (Reading database ... 5% (Reading database ...
10% (Reading database ... 15% (Reading database ... 20% (Reading
database ... 25% (Reading database ... 30% (Reading database ... 35%
(Reading database ... 40% (Reading database ... 45% (Reading
```

```
database ... 50% (Reading database ... 55% (Reading database ... 60%
(Reading database ... 65% (Reading database ... 70% (Reading
database ... 75% (Reading database ... 80% (Reading database ... 85%
(Reading database ... 90% (Reading database ... 95% (Reading
database ... 100% (Reading database ... 156213 files and directories
currently installed.)
Preparing to unpack .../libnvinfer5_5.1.5-1+cuda10.1_amd64.deb ...
Unpacking libnvinfer5 (5.1.5-1+cuda10.1) ...
Setting up libnvinfer5 (5.1.5-1+cuda10.1) ...
Processing triggers for libc-bin (2.27-3ubuntu1.3) ...
/sbin/ldconfig.real:
/usr/local/lib/python3.7/dist-packages/ideep4py/lib/libmkldnn.so.0 is
not a symbolic link


--2022-04-09 07:20:17--
https://developer.download.nvidia.com/compute/machine-learning/repos/
ubuntu1804/x86_64/nvidia-machine-learning-repo-ubuntu1804_1.0.0-
1_amd64.deb
Resolving developer.download.nvidia.com
(developer.download.nvidia.com)... 152.199.39.144
Connecting to developer.download.nvidia.com
(developer.download.nvidia.com)|152.199.39.144|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 2926 (2.9K) [application/x-deb]
Saving to: ‘nvidia-machine-learning-repo-ubuntu1804_1.0.0-1_amd64.deb’

     0K ..                                                    100%
132M=0s

2022-04-09 07:20:17 (132 MB/s) - ‘nvidia-machine-learning-repo-
ubuntu1804_1.0.0-1_amd64.deb’ saved [2926/2926]

W: Target Packages (Packages) is configured multiple times in
/etc/apt/sources.list.d/nvidia-machine-learning.list:1 and
/etc/apt/sources.list.d/nvidia-ml.list:1
W: Target Packages (Packages) is configured multiple times in
/etc/apt/sources.list.d/nvidia-machine-learning.list:1 and
/etc/apt/sources.list.d/nvidia-ml.list:1
debconf: unable to initialize frontend: Dialog
debconf: (No usable dialog-like program is installed, so the dialog
based frontend cannot be used. at
/usr/share/perl5/Debconf/FrontEnd/Dialog.pm line 76, <> line 1.)
debconf: falling back to frontend: Readline
debconf: unable to initialize frontend: Readline
debconf: (This frontend requires a controlling tty.)
debconf: falling back to frontend: Teletype
dpkg-preconfigure: unable to re-open stdin:
W: Target Packages (Packages) is configured multiple times in
```

```
/etc/apt/sources.list.d/nvidia-machine-learning.list:1 and
/etc/apt/sources.list.d/nvidia-ml.list:1

# check TensorRT version
print("TensorRT version: ")
!dpkg -l | grep nvinfer

TensorRT version:
ii  libnvinfer5                              5.1.5-1+cuda10.1
amd64        TensorRT runtime libraries
```

A successfull TensorRT installation looks like:

```
TensorRT version:
ii  libnvinfer5   5.1.5-1+cuda10.1   amd64        TensorRT runtime
libraries
```

Check Tensor core GPU

The below code check whether a Tensor-core GPU is present.

```
from tensorflow.python.client import device_lib

def check_tensor_core_gpu_present():
    local_device_protos = device_lib.list_local_devices()
    for line in local_device_protos:
        if "compute capability" in str(line):
            compute_capability =
float(line.physical_device_desc.split("compute capability: ")[-1])
            if compute_capability>=7.0:
                return True

print("Tensor Core GPU Present:", check_tensor_core_gpu_present())
tensor_core_gpu = check_tensor_core_gpu_present()

Tensor Core GPU Present: None
```

Importing required libraries

```
from __future__ import absolute_import, division, print_function,
unicode_literals
import os
import time

import numpy as np
import matplotlib.pyplot as plt

import tensorflow as tf
from tensorflow import keras
from tensorflow.python.compiler.tensorrt import trt_convert as trt
from tensorflow.python.saved_model import tag_constants
from tensorflow.keras.preprocessing import image
```

```
from tensorflow.keras.applications.inception_v3 import
InceptionV3,preprocess_input, decode_predictions

image_dir = '/content/drive/MyDrive/Colab Notebooks/train/Black-grass'

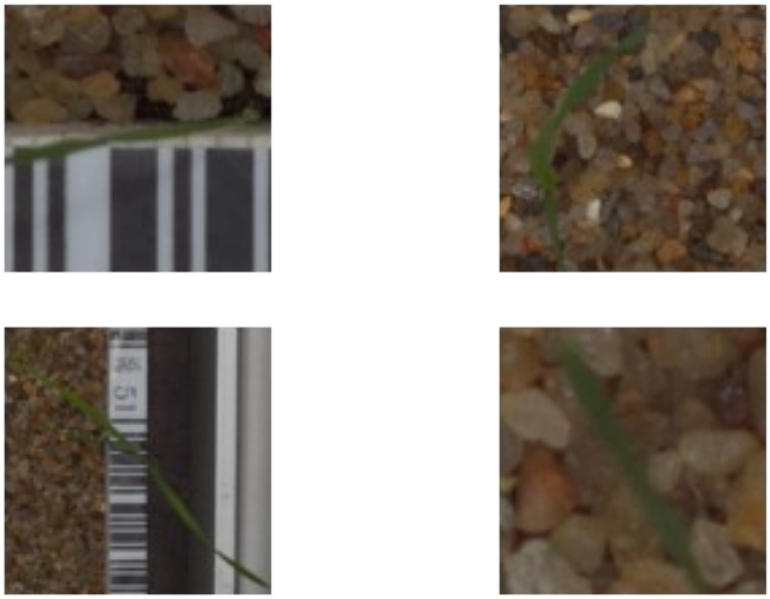
import os
images = os.listdir(image_dir)
images[1]
```

```
{"type":"string"}
```

```
from tensorflow.keras.preprocessing import image

fig, axes = plt.subplots(nrows=2, ncols=2)

for i in range(4):
  img_file = images[i]
  img_path = os.path.join(image_dir,img_file)
  img = image.load_img(img_path, target_size=(224, 224))
  plt.subplot(2,2,i+1)
  plt.imshow(img);
  plt.axis('off');
```



Model

We next download and test a Inception pre-trained model from the Keras model zoo.

```
inc_model = InceptionV3(weights = 'imagenet',include_top=False)

# inc_model.summary()
```

```
inc_model.trainable = False

x = inc_model.output

x = tf.keras.layers.Dense(1000,activation = 'softmax')(x)
model = tf.keras.models.Model(inputs = inc_model.input,outputs = x)
model.summary()
```

```
Model: "model"
__________________________________________________________________________________________________
Layer (type)                    Output Shape         Param #
Connected to
==================================================================================================
input_1 (InputLayer)            [(None, None, None,  0


__________________________________________________________________________________________________
conv2d (Conv2D)                 (None, None, None, 3 864
input_1[0][0]
__________________________________________________________________________________________________
batch_normalization (BatchNorma (None, None, None, 3 96
conv2d[0][0]
__________________________________________________________________________________________________
activation (Activation)         (None, None, None, 3 0
batch_normalization[0][0]
__________________________________________________________________________________________________
conv2d_1 (Conv2D)               (None, None, None, 3 9216
activation[0][0]
__________________________________________________________________________________________________
batch_normalization_1 (BatchNor (None, None, None, 3 96
conv2d_1[0][0]
__________________________________________________________________________________________________
activation_1 (Activation)       (None, None, None, 3 0
batch_normalization_1[0][0]
__________________________________________________________________________________________________
conv2d_2 (Conv2D)               (None, None, None, 6 18432
activation_1[0][0]
__________________________________________________________________________________________________
batch_normalization_2 (BatchNor (None, None, None, 6 192
conv2d_2[0][0]
__________________________________________________________________________________________________
```

```
_____
activation_2 (Activation)       (None, None, None, 6 0
batch_normalization_2[0][0]
_____
_____
max_pooling2d (MaxPooling2D)    (None, None, None, 6 0
activation_2[0][0]
_____
_____
conv2d_3 (Conv2D)               (None, None, None, 8 5120
max_pooling2d[0][0]
_____
_____
batch_normalization_3 (BatchNor (None, None, None, 8 240
conv2d_3[0][0]
_____
_____
activation_3 (Activation)       (None, None, None, 8 0
batch_normalization_3[0][0]
_____
_____
conv2d_4 (Conv2D)               (None, None, None, 1 138240
activation_3[0][0]
_____
_____
batch_normalization_4 (BatchNor (None, None, None, 1 576
conv2d_4[0][0]
_____
_____
activation_4 (Activation)       (None, None, None, 1 0
batch_normalization_4[0][0]
_____
_____
max_pooling2d_1 (MaxPooling2D)  (None, None, None, 1 0
activation_4[0][0]
_____
_____
conv2d_8 (Conv2D)               (None, None, None, 6 12288
max_pooling2d_1[0][0]
_____
_____
batch_normalization_8 (BatchNor (None, None, None, 6 192
conv2d_8[0][0]
_____
_____
activation_8 (Activation)       (None, None, None, 6 0
batch_normalization_8[0][0]
_____
_____
conv2d_6 (Conv2D)               (None, None, None, 4 9216
```

```
max_pooling2d_1[0][0]
__________________________________________________________________________________________________
conv2d_9 (Conv2D)               (None, None, None, 9 55296
activation_8[0][0]
__________________________________________________________________________________________________
batch_normalization_6 (BatchNor (None, None, None, 4 144
conv2d_6[0][0]
__________________________________________________________________________________________________
batch_normalization_9 (BatchNor (None, None, None, 9 288
conv2d_9[0][0]
__________________________________________________________________________________________________
activation_6 (Activation)       (None, None, None, 4 0
batch_normalization_6[0][0]
__________________________________________________________________________________________________
activation_9 (Activation)       (None, None, None, 9 0
batch_normalization_9[0][0]
__________________________________________________________________________________________________
average_pooling2d (AveragePooli (None, None, None, 1 0
max_pooling2d_1[0][0]
__________________________________________________________________________________________________
conv2d_5 (Conv2D)               (None, None, None, 6 12288
max_pooling2d_1[0][0]
__________________________________________________________________________________________________
conv2d_7 (Conv2D)               (None, None, None, 6 76800
activation_6[0][0]
__________________________________________________________________________________________________
conv2d_10 (Conv2D)              (None, None, None, 9 82944
activation_9[0][0]
__________________________________________________________________________________________________
conv2d_11 (Conv2D)              (None, None, None, 3 6144
average_pooling2d[0][0]
__________________________________________________________________________________________________
batch_normalization_5 (BatchNor (None, None, None, 6 192
conv2d_5[0][0]
__________________________________________________________________________________________________
batch_normalization_7 (BatchNor (None, None, None, 6 192
conv2d_7[0][0]
__________________________________________________________________________________________________
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_____________________________
batch_normalization_10 (BatchNo (None, None, None, 9 288
conv2d_10[0][0]
_____________________________________________________________________________
_____________________________
batch_normalization_11 (BatchNo (None, None, None, 3 96
conv2d_11[0][0]
_____________________________________________________________________________
_____________________________
activation_5 (Activation)       (None, None, None, 6 0
batch_normalization_5[0][0]
_____________________________________________________________________________
_____________________________
activation_7 (Activation)       (None, None, None, 6 0
batch_normalization_7[0][0]
_____________________________________________________________________________
_____________________________
activation_10 (Activation)      (None, None, None, 9 0
batch_normalization_10[0][0]
_____________________________________________________________________________
_____________________________
activation_11 (Activation)      (None, None, None, 3 0
batch_normalization_11[0][0]
_____________________________________________________________________________
_____________________________
mixed0 (Concatenate)            (None, None, None, 2 0
activation_5[0][0]

activation_7[0][0]

activation_10[0][0]

activation_11[0][0]
_____________________________________________________________________________
_____________________________
conv2d_15 (Conv2D)              (None, None, None, 6 16384
mixed0[0][0]
_____________________________________________________________________________
_____________________________
batch_normalization_15 (BatchNo (None, None, None, 6 192
conv2d_15[0][0]
_____________________________________________________________________________
_____________________________
activation_15 (Activation)      (None, None, None, 6 0
batch_normalization_15[0][0]
_____________________________________________________________________________
_____________________________
conv2d_13 (Conv2D)              (None, None, None, 4 12288
mixed0[0][0]
_____________________________________________________________________________
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_____
conv2d_16 (Conv2D)              (None, None, None, 9 55296
activation_15[0][0]
_____
_____
batch_normalization_13 (BatchNo (None, None, None, 4 144
conv2d_13[0][0]
_____
_____
batch_normalization_16 (BatchNo (None, None, None, 9 288
conv2d_16[0][0]
_____
_____
activation_13 (Activation)      (None, None, None, 4 0
batch_normalization_13[0][0]
_____
_____
activation_16 (Activation)      (None, None, None, 9 0
batch_normalization_16[0][0]
_____
_____
average_pooling2d_1 (AveragePoo (None, None, None, 2 0
mixed0[0][0]
_____
_____
conv2d_12 (Conv2D)              (None, None, None, 6 16384
mixed0[0][0]
_____
_____
conv2d_14 (Conv2D)              (None, None, None, 6 76800
activation_13[0][0]
_____
_____
conv2d_17 (Conv2D)              (None, None, None, 9 82944
activation_16[0][0]
_____
_____
conv2d_18 (Conv2D)              (None, None, None, 6 16384
average_pooling2d_1[0][0]
_____
_____
batch_normalization_12 (BatchNo (None, None, None, 6 192
conv2d_12[0][0]
_____
_____
batch_normalization_14 (BatchNo (None, None, None, 6 192
conv2d_14[0][0]
_____
_____
batch_normalization_17 (BatchNo (None, None, None, 9 288
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conv2d_17[0][0]
__________________________________________________________________________________________
__________________________
batch_normalization_18 (BatchNo (None, None, None, 6 192
conv2d_18[0][0]
__________________________________________________________________________________________
__________________________
activation_12 (Activation)      (None, None, None, 6 0
batch_normalization_12[0][0]
__________________________________________________________________________________________
__________________________
activation_14 (Activation)      (None, None, None, 6 0
batch_normalization_14[0][0]
__________________________________________________________________________________________
__________________________
activation_17 (Activation)      (None, None, None, 9 0
batch_normalization_17[0][0]
__________________________________________________________________________________________
__________________________
activation_18 (Activation)      (None, None, None, 6 0
batch_normalization_18[0][0]
__________________________________________________________________________________________
__________________________
mixed1 (Concatenate)            (None, None, None, 2 0
activation_12[0][0]

activation_14[0][0]

activation_17[0][0]

activation_18[0][0]
__________________________________________________________________________________________
__________________________
conv2d_22 (Conv2D)              (None, None, None, 6 18432
mixed1[0][0]
__________________________________________________________________________________________
__________________________
batch_normalization_22 (BatchNo (None, None, None, 6 192
conv2d_22[0][0]
__________________________________________________________________________________________
__________________________
activation_22 (Activation)      (None, None, None, 6 0
batch_normalization_22[0][0]
__________________________________________________________________________________________
__________________________
conv2d_20 (Conv2D)              (None, None, None, 4 13824
mixed1[0][0]
__________________________________________________________________________________________
__________________________
conv2d_23 (Conv2D)              (None, None, None, 9 55296
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activation_22[0][0]
__________________________________________________________________________
__________________________
batch_normalization_20 (BatchNo (None, None, None, 4 144
conv2d_20[0][0]
__________________________________________________________________________
__________________________
batch_normalization_23 (BatchNo (None, None, None, 9 288
conv2d_23[0][0]
__________________________________________________________________________
__________________________
activation_20 (Activation)      (None, None, None, 4 0
batch_normalization_20[0][0]
__________________________________________________________________________
__________________________
activation_23 (Activation)      (None, None, None, 9 0
batch_normalization_23[0][0]
__________________________________________________________________________
__________________________
average_pooling2d_2 (AveragePoo (None, None, None, 2 0
mixed1[0][0]
__________________________________________________________________________
__________________________
conv2d_19 (Conv2D)              (None, None, None, 6 18432
mixed1[0][0]
__________________________________________________________________________
__________________________
conv2d_21 (Conv2D)              (None, None, None, 6 76800
activation_20[0][0]
__________________________________________________________________________
__________________________
conv2d_24 (Conv2D)              (None, None, None, 9 82944
activation_23[0][0]
__________________________________________________________________________
__________________________
conv2d_25 (Conv2D)              (None, None, None, 6 18432
average_pooling2d_2[0][0]
__________________________________________________________________________
__________________________
batch_normalization_19 (BatchNo (None, None, None, 6 192
conv2d_19[0][0]
__________________________________________________________________________
__________________________
batch_normalization_21 (BatchNo (None, None, None, 6 192
conv2d_21[0][0]
__________________________________________________________________________
__________________________
batch_normalization_24 (BatchNo (None, None, None, 9 288
conv2d_24[0][0]
__________________________________________________________________________
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__________________________________
batch_normalization_25 (BatchNo (None, None, None, 6 192
conv2d_25[0][0]
__________________________________________________________________________________
__________________________________
activation_19 (Activation)      (None, None, None, 6 0
batch_normalization_19[0][0]
__________________________________________________________________________________
__________________________________
activation_21 (Activation)      (None, None, None, 6 0
batch_normalization_21[0][0]
__________________________________________________________________________________
__________________________________
activation_24 (Activation)      (None, None, None, 9 0
batch_normalization_24[0][0]
__________________________________________________________________________________
__________________________________
activation_25 (Activation)      (None, None, None, 6 0
batch_normalization_25[0][0]
__________________________________________________________________________________
__________________________________
mixed2 (Concatenate)            (None, None, None, 2 0
activation_19[0][0]

activation_21[0][0]

activation_24[0][0]

activation_25[0][0]
__________________________________________________________________________________
__________________________________
conv2d_27 (Conv2D)              (None, None, None, 6 18432
mixed2[0][0]
__________________________________________________________________________________
__________________________________
batch_normalization_27 (BatchNo (None, None, None, 6 192
conv2d_27[0][0]
__________________________________________________________________________________
__________________________________
activation_27 (Activation)      (None, None, None, 6 0
batch_normalization_27[0][0]
__________________________________________________________________________________
__________________________________
conv2d_28 (Conv2D)              (None, None, None, 9 55296
activation_27[0][0]
__________________________________________________________________________________
__________________________________
batch_normalization_28 (BatchNo (None, None, None, 9 288
conv2d_28[0][0]
__________________________________________________________________________________
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__________________________________
activation_28 (Activation)      (None, None, None, 9 0
batch_normalization_28[0][0]
__________________________________________________________________________________
__________________________________
conv2d_26 (Conv2D)              (None, None, None, 3 995328
mixed2[0][0]
__________________________________________________________________________________
__________________________________
conv2d_29 (Conv2D)              (None, None, None, 9 82944
activation_28[0][0]
__________________________________________________________________________________
__________________________________
batch_normalization_26 (BatchNo (None, None, None, 3 1152
conv2d_26[0][0]
__________________________________________________________________________________
__________________________________
batch_normalization_29 (BatchNo (None, None, None, 9 288
conv2d_29[0][0]
__________________________________________________________________________________
__________________________________
activation_26 (Activation)      (None, None, None, 3 0
batch_normalization_26[0][0]
__________________________________________________________________________________
__________________________________
activation_29 (Activation)      (None, None, None, 9 0
batch_normalization_29[0][0]
__________________________________________________________________________________
__________________________________
max_pooling2d_2 (MaxPooling2D)  (None, None, None, 2 0
mixed2[0][0]
__________________________________________________________________________________
__________________________________
mixed3 (Concatenate)            (None, None, None, 7 0
activation_26[0][0]

activation_29[0][0]

max_pooling2d_2[0][0]
__________________________________________________________________________________
__________________________________
conv2d_34 (Conv2D)              (None, None, None, 1 98304
mixed3[0][0]
__________________________________________________________________________________
__________________________________
batch_normalization_34 (BatchNo (None, None, None, 1 384
conv2d_34[0][0]
__________________________________________________________________________________
__________________________________
activation_34 (Activation)      (None, None, None, 1 0
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batch_normalization_34[0][0]
__________________________________________________________________________________
__________________________________
conv2d_35 (Conv2D)              (None, None, None, 1 114688
activation_34[0][0]
__________________________________________________________________________________
__________________________________
batch_normalization_35 (BatchNo (None, None, None, 1 384
conv2d_35[0][0]
__________________________________________________________________________________
__________________________________
activation_35 (Activation)      (None, None, None, 1 0
batch_normalization_35[0][0]
__________________________________________________________________________________
__________________________________
conv2d_31 (Conv2D)              (None, None, None, 1 98304
mixed3[0][0]
__________________________________________________________________________________
__________________________________
conv2d_36 (Conv2D)              (None, None, None, 1 114688
activation_35[0][0]
__________________________________________________________________________________
__________________________________
batch_normalization_31 (BatchNo (None, None, None, 1 384
conv2d_31[0][0]
__________________________________________________________________________________
__________________________________
batch_normalization_36 (BatchNo (None, None, None, 1 384
conv2d_36[0][0]
__________________________________________________________________________________
__________________________________
activation_31 (Activation)      (None, None, None, 1 0
batch_normalization_31[0][0]
__________________________________________________________________________________
__________________________________
activation_36 (Activation)      (None, None, None, 1 0
batch_normalization_36[0][0]
__________________________________________________________________________________
__________________________________
conv2d_32 (Conv2D)              (None, None, None, 1 114688
activation_31[0][0]
__________________________________________________________________________________
__________________________________
conv2d_37 (Conv2D)              (None, None, None, 1 114688
activation_36[0][0]
__________________________________________________________________________________
__________________________________
batch_normalization_32 (BatchNo (None, None, None, 1 384
conv2d_32[0][0]
__________________________________________________________________________________
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__________________________________
batch_normalization_37 (BatchNo (None, None, None, 1 384
conv2d_37[0][0]
__________________________________________________________________________________
__________________________________
activation_32 (Activation)      (None, None, None, 1 0
batch_normalization_32[0][0]
__________________________________________________________________________________
__________________________________
activation_37 (Activation)      (None, None, None, 1 0
batch_normalization_37[0][0]
__________________________________________________________________________________
__________________________________
average_pooling2d_3 (AveragePoo (None, None, None, 7 0
mixed3[0][0]
__________________________________________________________________________________
__________________________________
conv2d_30 (Conv2D)              (None, None, None, 1 147456
mixed3[0][0]
__________________________________________________________________________________
__________________________________
conv2d_33 (Conv2D)              (None, None, None, 1 172032
activation_32[0][0]
__________________________________________________________________________________
__________________________________
conv2d_38 (Conv2D)              (None, None, None, 1 172032
activation_37[0][0]
__________________________________________________________________________________
__________________________________
conv2d_39 (Conv2D)              (None, None, None, 1 147456
average_pooling2d_3[0][0]
__________________________________________________________________________________
__________________________________
batch_normalization_30 (BatchNo (None, None, None, 1 576
conv2d_30[0][0]
__________________________________________________________________________________
__________________________________
batch_normalization_33 (BatchNo (None, None, None, 1 576
conv2d_33[0][0]
__________________________________________________________________________________
__________________________________
batch_normalization_38 (BatchNo (None, None, None, 1 576
conv2d_38[0][0]
__________________________________________________________________________________
__________________________________
batch_normalization_39 (BatchNo (None, None, None, 1 576
conv2d_39[0][0]
__________________________________________________________________________________
__________________________________
activation_30 (Activation)      (None, None, None, 1 0
```

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batch_normalization_30[0][0]
__________________________________________________________________________________
__________________________________
activation_33 (Activation)      (None, None, None, 1 0
batch_normalization_33[0][0]
__________________________________________________________________________________
__________________________________
activation_38 (Activation)      (None, None, None, 1 0
batch_normalization_38[0][0]
__________________________________________________________________________________
__________________________________
activation_39 (Activation)      (None, None, None, 1 0
batch_normalization_39[0][0]
__________________________________________________________________________________
__________________________________
mixed4 (Concatenate)            (None, None, None, 7 0
activation_30[0][0]

activation_33[0][0]

activation_38[0][0]

activation_39[0][0]
__________________________________________________________________________________
__________________________________
conv2d_44 (Conv2D)              (None, None, None, 1 122880
mixed4[0][0]
__________________________________________________________________________________
__________________________________
batch_normalization_44 (BatchNo (None, None, None, 1 480
conv2d_44[0][0]
__________________________________________________________________________________
__________________________________
activation_44 (Activation)      (None, None, None, 1 0
batch_normalization_44[0][0]
__________________________________________________________________________________
__________________________________
conv2d_45 (Conv2D)              (None, None, None, 1 179200
activation_44[0][0]
__________________________________________________________________________________
__________________________________
batch_normalization_45 (BatchNo (None, None, None, 1 480
conv2d_45[0][0]
__________________________________________________________________________________
__________________________________
activation_45 (Activation)      (None, None, None, 1 0
batch_normalization_45[0][0]
__________________________________________________________________________________
__________________________________
conv2d_41 (Conv2D)              (None, None, None, 1 122880
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mixed4[0][0]
__________________________________________________________________________________________________
__________________________________
conv2d_46 (Conv2D)              (None, None, None, 1 179200
activation_45[0][0]
__________________________________________________________________________________________________
__________________________________
batch_normalization_41 (BatchNo (None, None, None, 1 480
conv2d_41[0][0]
__________________________________________________________________________________________________
__________________________________
batch_normalization_46 (BatchNo (None, None, None, 1 480
conv2d_46[0][0]
__________________________________________________________________________________________________
__________________________________
activation_41 (Activation)      (None, None, None, 1 0
batch_normalization_41[0][0]
__________________________________________________________________________________________________
__________________________________
activation_46 (Activation)      (None, None, None, 1 0
batch_normalization_46[0][0]
__________________________________________________________________________________________________
__________________________________
conv2d_42 (Conv2D)              (None, None, None, 1 179200
activation_41[0][0]
__________________________________________________________________________________________________
__________________________________
conv2d_47 (Conv2D)              (None, None, None, 1 179200
activation_46[0][0]
__________________________________________________________________________________________________
__________________________________
batch_normalization_42 (BatchNo (None, None, None, 1 480
conv2d_42[0][0]
__________________________________________________________________________________________________
__________________________________
batch_normalization_47 (BatchNo (None, None, None, 1 480
conv2d_47[0][0]
__________________________________________________________________________________________________
__________________________________
activation_42 (Activation)      (None, None, None, 1 0
batch_normalization_42[0][0]
__________________________________________________________________________________________________
__________________________________
activation_47 (Activation)      (None, None, None, 1 0
batch_normalization_47[0][0]
__________________________________________________________________________________________________
__________________________________
average_pooling2d_4 (AveragePoo (None, None, None, 7 0
mixed4[0][0]
__________________________________________________________________________________________________
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__________________________________
conv2d_40 (Conv2D)              (None, None, None, 1 147456
mixed4[0][0]
__________________________________________________________________________________
__________________________________
conv2d_43 (Conv2D)              (None, None, None, 1 215040
activation_42[0][0]
__________________________________________________________________________________
__________________________________
conv2d_48 (Conv2D)              (None, None, None, 1 215040
activation_47[0][0]
__________________________________________________________________________________
__________________________________
conv2d_49 (Conv2D)              (None, None, None, 1 147456
average_pooling2d_4[0][0]
__________________________________________________________________________________
__________________________________
batch_normalization_40 (BatchNo (None, None, None, 1 576
conv2d_40[0][0]
__________________________________________________________________________________
__________________________________
batch_normalization_43 (BatchNo (None, None, None, 1 576
conv2d_43[0][0]
__________________________________________________________________________________
__________________________________
batch_normalization_48 (BatchNo (None, None, None, 1 576
conv2d_48[0][0]
__________________________________________________________________________________
__________________________________
batch_normalization_49 (BatchNo (None, None, None, 1 576
conv2d_49[0][0]
__________________________________________________________________________________
__________________________________
activation_40 (Activation)      (None, None, None, 1 0
batch_normalization_40[0][0]
__________________________________________________________________________________
__________________________________
activation_43 (Activation)      (None, None, None, 1 0
batch_normalization_43[0][0]
__________________________________________________________________________________
__________________________________
activation_48 (Activation)      (None, None, None, 1 0
batch_normalization_48[0][0]
__________________________________________________________________________________
__________________________________
activation_49 (Activation)      (None, None, None, 1 0
batch_normalization_49[0][0]
__________________________________________________________________________________
__________________________________
mixed5 (Concatenate)            (None, None, None, 7 0
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activation_40[0][0]

activation_43[0][0]

activation_48[0][0]

activation_49[0][0]
__________________________________________________________________________________________________
conv2d_54 (Conv2D)              (None, None, None, 1 122880
mixed5[0][0]
__________________________________________________________________________________________________
batch_normalization_54 (BatchNo (None, None, None, 1 480
conv2d_54[0][0]
__________________________________________________________________________________________________
activation_54 (Activation)      (None, None, None, 1 0
batch_normalization_54[0][0]
__________________________________________________________________________________________________
conv2d_55 (Conv2D)              (None, None, None, 1 179200
activation_54[0][0]
__________________________________________________________________________________________________
batch_normalization_55 (BatchNo (None, None, None, 1 480
conv2d_55[0][0]
__________________________________________________________________________________________________
activation_55 (Activation)      (None, None, None, 1 0
batch_normalization_55[0][0]
__________________________________________________________________________________________________
conv2d_51 (Conv2D)              (None, None, None, 1 122880
mixed5[0][0]
__________________________________________________________________________________________________
conv2d_56 (Conv2D)              (None, None, None, 1 179200
activation_55[0][0]
__________________________________________________________________________________________________
batch_normalization_51 (BatchNo (None, None, None, 1 480
conv2d_51[0][0]
__________________________________________________________________________________________________
batch_normalization_56 (BatchNo (None, None, None, 1 480
conv2d_56[0][0]
__________________________________________________________________________________________________
activation_51 (Activation)      (None, None, None, 1 0
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batch_normalization_51[0][0]
__________________________________________________________________________________
__________________________________
activation_56 (Activation)      (None, None, None, 1 0
batch_normalization_56[0][0]
__________________________________________________________________________________
__________________________________
conv2d_52 (Conv2D)              (None, None, None, 1 179200
activation_51[0][0]
__________________________________________________________________________________
__________________________________
conv2d_57 (Conv2D)              (None, None, None, 1 179200
activation_56[0][0]
__________________________________________________________________________________
__________________________________
batch_normalization_52 (BatchNo (None, None, None, 1 480
conv2d_52[0][0]
__________________________________________________________________________________
__________________________________
batch_normalization_57 (BatchNo (None, None, None, 1 480
conv2d_57[0][0]
__________________________________________________________________________________
__________________________________
activation_52 (Activation)      (None, None, None, 1 0
batch_normalization_52[0][0]
__________________________________________________________________________________
__________________________________
activation_57 (Activation)      (None, None, None, 1 0
batch_normalization_57[0][0]
__________________________________________________________________________________
__________________________________
average_pooling2d_5 (AveragePoo (None, None, None, 7 0
mixed5[0][0]
__________________________________________________________________________________
__________________________________
conv2d_50 (Conv2D)              (None, None, None, 1 147456
mixed5[0][0]
__________________________________________________________________________________
__________________________________
conv2d_53 (Conv2D)              (None, None, None, 1 215040
activation_52[0][0]
__________________________________________________________________________________
__________________________________
conv2d_58 (Conv2D)              (None, None, None, 1 215040
activation_57[0][0]
__________________________________________________________________________________
__________________________________
conv2d_59 (Conv2D)              (None, None, None, 1 147456
average_pooling2d_5[0][0]
__________________________________________________________________________________
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__________________________________
batch_normalization_50 (BatchNo (None, None, None, 1 576
conv2d_50[0][0]
__________________________________________________________________________________
__________________________________
batch_normalization_53 (BatchNo (None, None, None, 1 576
conv2d_53[0][0]
__________________________________________________________________________________
__________________________________
batch_normalization_58 (BatchNo (None, None, None, 1 576
conv2d_58[0][0]
__________________________________________________________________________________
__________________________________
batch_normalization_59 (BatchNo (None, None, None, 1 576
conv2d_59[0][0]
__________________________________________________________________________________
__________________________________
activation_50 (Activation)      (None, None, None, 1 0
batch_normalization_50[0][0]
__________________________________________________________________________________
__________________________________
activation_53 (Activation)      (None, None, None, 1 0
batch_normalization_53[0][0]
__________________________________________________________________________________
__________________________________
activation_58 (Activation)      (None, None, None, 1 0
batch_normalization_58[0][0]
__________________________________________________________________________________
__________________________________
activation_59 (Activation)      (None, None, None, 1 0
batch_normalization_59[0][0]
__________________________________________________________________________________
__________________________________
mixed6 (Concatenate)            (None, None, None, 7 0
activation_50[0][0]

activation_53[0][0]

activation_58[0][0]

activation_59[0][0]
__________________________________________________________________________________
__________________________________
conv2d_64 (Conv2D)              (None, None, None, 1 147456
mixed6[0][0]
__________________________________________________________________________________
__________________________________
batch_normalization_64 (BatchNo (None, None, None, 1 576
conv2d_64[0][0]
__________________________________________________________________________________
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_____
activation_64 (Activation)      (None, None, None, 1 0
batch_normalization_64[0][0]
_____
_____
conv2d_65 (Conv2D)              (None, None, None, 1 258048
activation_64[0][0]
_____
_____
batch_normalization_65 (BatchNo (None, None, None, 1 576
conv2d_65[0][0]
_____
_____
activation_65 (Activation)      (None, None, None, 1 0
batch_normalization_65[0][0]
_____
_____
conv2d_61 (Conv2D)              (None, None, None, 1 147456
mixed6[0][0]
_____
_____
conv2d_66 (Conv2D)              (None, None, None, 1 258048
activation_65[0][0]
_____
_____
batch_normalization_61 (BatchNo (None, None, None, 1 576
conv2d_61[0][0]
_____
_____
batch_normalization_66 (BatchNo (None, None, None, 1 576
conv2d_66[0][0]
_____
_____
activation_61 (Activation)      (None, None, None, 1 0
batch_normalization_61[0][0]
_____
_____
activation_66 (Activation)      (None, None, None, 1 0
batch_normalization_66[0][0]
_____
_____
conv2d_62 (Conv2D)              (None, None, None, 1 258048
activation_61[0][0]
_____
_____
conv2d_67 (Conv2D)              (None, None, None, 1 258048
activation_66[0][0]
_____
_____
batch_normalization_62 (BatchNo (None, None, None, 1 576
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conv2d_62[0][0]
__________________________________________________________________________________________________
__________________________
batch_normalization_67 (BatchNo (None, None, None, 1 576
conv2d_67[0][0]
__________________________________________________________________________________________________
__________________________
activation_62 (Activation)      (None, None, None, 1 0
batch_normalization_62[0][0]
__________________________________________________________________________________________________
__________________________
activation_67 (Activation)      (None, None, None, 1 0
batch_normalization_67[0][0]
__________________________________________________________________________________________________
__________________________
average_pooling2d_6 (AveragePoo (None, None, None, 7 0
mixed6[0][0]
__________________________________________________________________________________________________
__________________________
conv2d_60 (Conv2D)              (None, None, None, 1 147456
mixed6[0][0]
__________________________________________________________________________________________________
__________________________
conv2d_63 (Conv2D)              (None, None, None, 1 258048
activation_62[0][0]
__________________________________________________________________________________________________
__________________________
conv2d_68 (Conv2D)              (None, None, None, 1 258048
activation_67[0][0]
__________________________________________________________________________________________________
__________________________
conv2d_69 (Conv2D)              (None, None, None, 1 147456
average_pooling2d_6[0][0]
__________________________________________________________________________________________________
__________________________
batch_normalization_60 (BatchNo (None, None, None, 1 576
conv2d_60[0][0]
__________________________________________________________________________________________________
__________________________
batch_normalization_63 (BatchNo (None, None, None, 1 576
conv2d_63[0][0]
__________________________________________________________________________________________________
__________________________
batch_normalization_68 (BatchNo (None, None, None, 1 576
conv2d_68[0][0]
__________________________________________________________________________________________________
__________________________
batch_normalization_69 (BatchNo (None, None, None, 1 576
conv2d_69[0][0]
__________________________________________________________________________________________________
```

```
__________________________________
activation_60 (Activation)      (None, None, None, 1 0
batch_normalization_60[0][0]
__________________________________________________________________________________________
__________________________________
activation_63 (Activation)      (None, None, None, 1 0
batch_normalization_63[0][0]
__________________________________________________________________________________________
__________________________________
activation_68 (Activation)      (None, None, None, 1 0
batch_normalization_68[0][0]
__________________________________________________________________________________________
__________________________________
activation_69 (Activation)      (None, None, None, 1 0
batch_normalization_69[0][0]
__________________________________________________________________________________________
__________________________________
mixed7 (Concatenate)            (None, None, None, 7 0
activation_60[0][0]

activation_63[0][0]

activation_68[0][0]

activation_69[0][0]
__________________________________________________________________________________________
__________________________________
conv2d_72 (Conv2D)              (None, None, None, 1 147456
mixed7[0][0]
__________________________________________________________________________________________
__________________________________
batch_normalization_72 (BatchNo (None, None, None, 1 576
conv2d_72[0][0]
__________________________________________________________________________________________
__________________________________
activation_72 (Activation)      (None, None, None, 1 0
batch_normalization_72[0][0]
__________________________________________________________________________________________
__________________________________
conv2d_73 (Conv2D)              (None, None, None, 1 258048
activation_72[0][0]
__________________________________________________________________________________________
__________________________________
batch_normalization_73 (BatchNo (None, None, None, 1 576
conv2d_73[0][0]
__________________________________________________________________________________________
__________________________________
activation_73 (Activation)      (None, None, None, 1 0
batch_normalization_73[0][0]
__________________________________________________________________________________________
```

```
_____
conv2d_70 (Conv2D)              (None, None, None, 1 147456
mixed7[0][0]
__________________________________________________________________________________
_____
conv2d_74 (Conv2D)              (None, None, None, 1 258048
activation_73[0][0]
__________________________________________________________________________________
_____
batch_normalization_70 (BatchNo (None, None, None, 1 576
conv2d_70[0][0]
__________________________________________________________________________________
_____
batch_normalization_74 (BatchNo (None, None, None, 1 576
conv2d_74[0][0]
__________________________________________________________________________________
_____
activation_70 (Activation)      (None, None, None, 1 0
batch_normalization_70[0][0]
__________________________________________________________________________________
_____
activation_74 (Activation)      (None, None, None, 1 0
batch_normalization_74[0][0]
__________________________________________________________________________________
_____
conv2d_71 (Conv2D)              (None, None, None, 3 552960
activation_70[0][0]
__________________________________________________________________________________
_____
conv2d_75 (Conv2D)              (None, None, None, 1 331776
activation_74[0][0]
__________________________________________________________________________________
_____
batch_normalization_71 (BatchNo (None, None, None, 3 960
conv2d_71[0][0]
__________________________________________________________________________________
_____
batch_normalization_75 (BatchNo (None, None, None, 1 576
conv2d_75[0][0]
__________________________________________________________________________________
_____
activation_71 (Activation)      (None, None, None, 3 0
batch_normalization_71[0][0]
__________________________________________________________________________________
_____
activation_75 (Activation)      (None, None, None, 1 0
batch_normalization_75[0][0]
__________________________________________________________________________________
_____
max_pooling2d_3 (MaxPooling2D)  (None, None, None, 7 0
```

```
mixed7[0][0]
__________________________________________________________________________________
__________________________
mixed8 (Concatenate)            (None, None, None, 1 0
activation_71[0][0]

activation_75[0][0]

max_pooling2d_3[0][0]
__________________________________________________________________________________
__________________________
conv2d_80 (Conv2D)              (None, None, None, 4 573440
mixed8[0][0]
__________________________________________________________________________________
__________________________
batch_normalization_80 (BatchNo (None, None, None, 4 1344
conv2d_80[0][0]
__________________________________________________________________________________
__________________________
activation_80 (Activation)      (None, None, None, 4 0
batch_normalization_80[0][0]
__________________________________________________________________________________
__________________________
conv2d_77 (Conv2D)              (None, None, None, 3 491520
mixed8[0][0]
__________________________________________________________________________________
__________________________
conv2d_81 (Conv2D)              (None, None, None, 3 1548288
activation_80[0][0]
__________________________________________________________________________________
__________________________
batch_normalization_77 (BatchNo (None, None, None, 3 1152
conv2d_77[0][0]
__________________________________________________________________________________
__________________________
batch_normalization_81 (BatchNo (None, None, None, 3 1152
conv2d_81[0][0]
__________________________________________________________________________________
__________________________
activation_77 (Activation)      (None, None, None, 3 0
batch_normalization_77[0][0]
__________________________________________________________________________________
__________________________
activation_81 (Activation)      (None, None, None, 3 0
batch_normalization_81[0][0]
__________________________________________________________________________________
__________________________
conv2d_78 (Conv2D)              (None, None, None, 3 442368
activation_77[0][0]
__________________________________________________________________________________
```

```
__________________________________
conv2d_79 (Conv2D)              (None, None, None, 3 442368
activation_77[0][0]
__________________________________________________________________________________
__________________________________
conv2d_82 (Conv2D)              (None, None, None, 3 442368
activation_81[0][0]
__________________________________________________________________________________
__________________________________
conv2d_83 (Conv2D)              (None, None, None, 3 442368
activation_81[0][0]
__________________________________________________________________________________
__________________________________
average_pooling2d_7 (AveragePoo (None, None, None, 1 0
mixed8[0][0]
__________________________________________________________________________________
__________________________________
conv2d_76 (Conv2D)              (None, None, None, 3 409600
mixed8[0][0]
__________________________________________________________________________________
__________________________________
batch_normalization_78 (BatchNo (None, None, None, 3 1152
conv2d_78[0][0]
__________________________________________________________________________________
__________________________________
batch_normalization_79 (BatchNo (None, None, None, 3 1152
conv2d_79[0][0]
__________________________________________________________________________________
__________________________________
batch_normalization_82 (BatchNo (None, None, None, 3 1152
conv2d_82[0][0]
__________________________________________________________________________________
__________________________________
batch_normalization_83 (BatchNo (None, None, None, 3 1152
conv2d_83[0][0]
__________________________________________________________________________________
__________________________________
conv2d_84 (Conv2D)              (None, None, None, 1 245760
average_pooling2d_7[0][0]
__________________________________________________________________________________
__________________________________
batch_normalization_76 (BatchNo (None, None, None, 3 960
conv2d_76[0][0]
__________________________________________________________________________________
__________________________________
activation_78 (Activation)      (None, None, None, 3 0
batch_normalization_78[0][0]
__________________________________________________________________________________
__________________________________
activation_79 (Activation)      (None, None, None, 3 0
```

```
batch_normalization_79[0][0]
__________________________________________________________________________________
_____________________________
activation_82 (Activation)      (None, None, None, 3 0
batch_normalization_82[0][0]
__________________________________________________________________________________
_____________________________
activation_83 (Activation)      (None, None, None, 3 0
batch_normalization_83[0][0]
__________________________________________________________________________________
_____________________________
batch_normalization_84 (BatchNo (None, None, None, 1 576
conv2d_84[0][0]
__________________________________________________________________________________
_____________________________
activation_76 (Activation)      (None, None, None, 3 0
batch_normalization_76[0][0]
__________________________________________________________________________________
_____________________________
mixed9_0 (Concatenate)          (None, None, None, 7 0
activation_78[0][0]

activation_79[0][0]
__________________________________________________________________________________
_____________________________
concatenate (Concatenate)       (None, None, None, 7 0
activation_82[0][0]

activation_83[0][0]
__________________________________________________________________________________
_____________________________
activation_84 (Activation)      (None, None, None, 1 0
batch_normalization_84[0][0]
__________________________________________________________________________________
_____________________________
mixed9 (Concatenate)            (None, None, None, 2 0
activation_76[0][0]

mixed9_0[0][0]

concatenate[0][0]

activation_84[0][0]
__________________________________________________________________________________
_____________________________
conv2d_89 (Conv2D)              (None, None, None, 4 917504
mixed9[0][0]
__________________________________________________________________________________
_____________________________
batch_normalization_89 (BatchNo (None, None, None, 4 1344
```

```
conv2d_89[0][0]
__________________________________________________________________________________________
__________________________
activation_89 (Activation)      (None, None, None, 4 0
batch_normalization_89[0][0]
__________________________________________________________________________________________
__________________________
conv2d_86 (Conv2D)              (None, None, None, 3 786432
mixed9[0][0]
__________________________________________________________________________________________
__________________________
conv2d_90 (Conv2D)              (None, None, None, 3 1548288
activation_89[0][0]
__________________________________________________________________________________________
__________________________
batch_normalization_86 (BatchNo (None, None, None, 3 1152
conv2d_86[0][0]
__________________________________________________________________________________________
__________________________
batch_normalization_90 (BatchNo (None, None, None, 3 1152
conv2d_90[0][0]
__________________________________________________________________________________________
__________________________
activation_86 (Activation)      (None, None, None, 3 0
batch_normalization_86[0][0]
__________________________________________________________________________________________
__________________________
activation_90 (Activation)      (None, None, None, 3 0
batch_normalization_90[0][0]
__________________________________________________________________________________________
__________________________
conv2d_87 (Conv2D)              (None, None, None, 3 442368
activation_86[0][0]
__________________________________________________________________________________________
__________________________
conv2d_88 (Conv2D)              (None, None, None, 3 442368
activation_86[0][0]
__________________________________________________________________________________________
__________________________
conv2d_91 (Conv2D)              (None, None, None, 3 442368
activation_90[0][0]
__________________________________________________________________________________________
__________________________
conv2d_92 (Conv2D)              (None, None, None, 3 442368
activation_90[0][0]
__________________________________________________________________________________________
__________________________
average_pooling2d_8 (AveragePoo (None, None, None, 2 0
mixed9[0][0]
__________________________________________________________________________________________
```

```
__________________________________
conv2d_85 (Conv2D)              (None, None, None, 3 655360
mixed9[0][0]
__________________________________________________________________________________
__________________________________
batch_normalization_87 (BatchNo (None, None, None, 3 1152
conv2d_87[0][0]
__________________________________________________________________________________
__________________________________
batch_normalization_88 (BatchNo (None, None, None, 3 1152
conv2d_88[0][0]
__________________________________________________________________________________
__________________________________
batch_normalization_91 (BatchNo (None, None, None, 3 1152
conv2d_91[0][0]
__________________________________________________________________________________
__________________________________
batch_normalization_92 (BatchNo (None, None, None, 3 1152
conv2d_92[0][0]
__________________________________________________________________________________
__________________________________
conv2d_93 (Conv2D)              (None, None, None, 1 393216
average_pooling2d_8[0][0]
__________________________________________________________________________________
__________________________________
batch_normalization_85 (BatchNo (None, None, None, 3 960
conv2d_85[0][0]
__________________________________________________________________________________
__________________________________
activation_87 (Activation)      (None, None, None, 3 0
batch_normalization_87[0][0]
__________________________________________________________________________________
__________________________________
activation_88 (Activation)      (None, None, None, 3 0
batch_normalization_88[0][0]
__________________________________________________________________________________
__________________________________
activation_91 (Activation)      (None, None, None, 3 0
batch_normalization_91[0][0]
__________________________________________________________________________________
__________________________________
activation_92 (Activation)      (None, None, None, 3 0
batch_normalization_92[0][0]
__________________________________________________________________________________
__________________________________
batch_normalization_93 (BatchNo (None, None, None, 1 576
conv2d_93[0][0]
__________________________________________________________________________________
__________________________________
activation_85 (Activation)      (None, None, None, 3 0
```

```
batch_normalization_85[0][0]
__________________________________________________________________________________
_____________________________
mixed9_1 (Concatenate)          (None, None, None, 7 0
activation_87[0][0]

activation_88[0][0]
__________________________________________________________________________________
_____________________________
concatenate_1 (Concatenate)     (None, None, None, 7 0
activation_91[0][0]

activation_92[0][0]
__________________________________________________________________________________
_____________________________
activation_93 (Activation)      (None, None, None, 1 0
batch_normalization_93[0][0]
__________________________________________________________________________________
_____________________________
mixed10 (Concatenate)           (None, None, None, 2 0
activation_85[0][0]

mixed9_1[0][0]

concatenate_1[0][0]

activation_93[0][0]
__________________________________________________________________________________
_____________________________
dense (Dense)                   (None, None, None, 1 2049000
mixed10[0][0]
==================================================================================
============================
Total params: 23,851,784
Trainable params: 2,049,000
Non-trainable params: 21,802,784
__________________________________________________________________________________
_____________________________
```

```python
for i in range(4):
  img_file = images[i]
  img_path = os.path.join(image_dir,img_file)
  img = image.load_img(img_path, target_size=(224, 224))
  x = image.img_to_array(img)
  x = np.expand_dims(x, axis=0)
  x = preprocess_input(x)
  print(x.shape)

  preds = model.predict(x)
  print(preds.shape)
```

```
  p = preds.reshape(-1,1000)
  print(p.shape)
  # decode the results into a list of tuples (class, description,
probability)
  # (one such list for each sample in the batch)


  #print('{} - Predicted: {}'.format(img_path,
decode_predictions(preds, top=3)[0]))
  print('{} - Predicted: {}'.format(img_path, decode_predictions(p,
top=3)[0]))

  plt.subplot(2,2,i+1)
  plt.imshow(img);
  plt.axis('off');
  plt.title(decode_predictions(p, top=3)[0][0][1])
  #plt.title(decode_predictions(preds, top=3)[0][0][1])
```

```
(1, 224, 224, 3)
(1, 5, 5, 1000)
(25, 1000)
Downloading data from
https://storage.googleapis.com/download.tensorflow.org/data/imagenet_c
lass_index.json
40960/35363 [==================================] - 0s 0us/step
49152/35363 [==========================================] - 0s 0us/step
/content/drive/MyDrive/Colab Notebooks/train/Black-grass/0050f38b3.png
- Predicted: [('n02093859', 'Kerry_blue_terrier', 0.04982188),
('n03770679', 'minivan', 0.027955458), ('n04429376', 'throne',
0.023945093)]
(1, 224, 224, 3)
(1, 5, 5, 1000)
(25, 1000)
/content/drive/MyDrive/Colab Notebooks/train/Black-grass/0183fdf68.png
- Predicted: [('n03483316', 'hand_blower', 0.010207782), ('n04483307',
'trimaran', 0.0073360736), ('n04597913', 'wooden_spoon', 0.007334963)]
(1, 224, 224, 3)
(1, 5, 5, 1000)
(25, 1000)
/content/drive/MyDrive/Colab Notebooks/train/Black-grass/0260cffa8.png
- Predicted: [('n02422106', 'hartebeest', 0.0043690717), ('n10565667',
'scuba_diver', 0.0036677658), ('n01847000', 'drake', 0.0033500409)]
(1, 224, 224, 3)
(1, 5, 5, 1000)
(25, 1000)
/content/drive/MyDrive/Colab Notebooks/train/Black-grass/05eedce4d.png
- Predicted: [('n07718472', 'cucumber', 0.0107887015), ('n09428293',
```

```
'seashore', 0.010220564), ('n02089973', 'English_foxhound',
0.008798744)]
```



TF-TRT takes input as aTensorFlow saved model, therefore, we re-export the Keras model as a TF saved model.

```
# Save the entire model as a SavedModel.
model.save('/content/drive/MyDrive/Colab Notebooks/inc_saved_model')
```

```
WARNING:tensorflow:From
/usr/local/lib/python3.7/dist-packages/tensorflow_core/python/ops/
resource_variable_ops.py:1781: calling BaseResourceVariable.__init__
(from tensorflow.python.ops.resource_variable_ops) with constraint is
deprecated and will be removed in a future version.
Instructions for updating:
If using Keras pass *_constraint arguments to layers.
INFO:tensorflow:Assets written to: /content/drive/MyDrive/Colab
Notebooks/inc_saved_model/assets
```

```
!saved_model_cli show --all --dir inc_saved_model
```

```
Traceback (most recent call last):
  File "/usr/local/bin/saved_model_cli", line 8, in <module>
    sys.exit(main())
  File
"/usr/local/lib/python3.7/dist-packages/tensorflow_core/python/tools/
saved_model_cli.py", line 909, in main
    args.func(args)
  File
"/usr/local/lib/python3.7/dist-packages/tensorflow_core/python/tools/
saved_model_cli.py", line 610, in show
```

```
    _show_all(args.dir)
  File
"/usr/local/lib/python3.7/dist-packages/tensorflow_core/python/tools/
saved_model_cli.py", line 192, in _show_all
    tag_sets =
saved_model_utils.get_saved_model_tag_sets(saved_model_dir)
  File
"/usr/local/lib/python3.7/dist-packages/tensorflow_core/python/tools/
saved_model_utils.py", line 88, in get_saved_model_tag_sets
    saved_model = read_saved_model(saved_model_dir)
  File
"/usr/local/lib/python3.7/dist-packages/tensorflow_core/python/tools/
saved_model_utils.py", line 55, in read_saved_model
    raise IOError("SavedModel file does not exist at: %s" %
saved_model_dir)
OSError: SavedModel file does not exist at: inc_saved_model
```

Inference with native TF2.0 saved model

```
model = tf.keras.models.load_model('/content/drive/MyDrive/Colab
Notebooks/inc_saved_model')

img_file = images[0]
img_path = os.path.join(image_dir,img_file)
img = image.load_img(img_path, target_size=(224, 224))
x = image.img_to_array(img)
x = np.expand_dims(x, axis=0)
x = preprocess_input(x)

preds = model.predict(x)

p = preds.reshape(-1,1000)
print(p.shape)


# decode the results into a list of tuples (class, description,
probability)
# (one such list for each sample in the batch)
#print('{} - Predicted: {}'.format(img_path, decode_predictions(preds,
top=3)[0]))
print('{} - Predicted: {}'.format(img_path, decode_predictions(p,
top=3)[0]))

plt.subplot(2,2,1)
plt.imshow(img);
plt.axis('off');
plt.title(decode_predictions(p, top=3)[0][0][1])
#plt.title(decode_predictions(preds, top=3)[0][0][1])

(25, 1000)
Downloading data from
```

```
https://storage.googleapis.com/download.tensorflow.org/data/imagenet_c
lass_index.json
40960/35363 [==================================] - 0s 0us/step
/content/drive/MyDrive/Colab Notebooks/train/Black-grass/0050f38b3.png
- Predicted: [('n07892512', 'red_wine', 0.05653725), ('n03110669',
'cornet', 0.041734356), ('n01729322', 'hognose_snake', 0.032864667)]

Text(0.5, 1.0, 'red_wine')
```




```
batch_size = 8
batched_input = np.zeros((batch_size, 224, 224, 3), dtype=np.float32)

for i in range(batch_size):
  img_file = images[i%4]
  img_path = os.path.join(image_dir,img_file)
  img = image.load_img(img_path, target_size=(224, 224))
  x = image.img_to_array(img)
  x = np.expand_dims(x, axis=0)
  x = preprocess_input(x)
  batched_input[i, :] = x
batched_input = tf.constant(batched_input)
print('batched_input shape: ', batched_input.shape)

batched_input shape:  (8, 224, 224, 3)

# Benchmarking throughput

N_run = 300
elapsed_time = []

for i in range(N_run):
  start_time = time.time()
  preds = model.predict(batched_input)
  end_time = time.time()
  elapsed_time = np.append(elapsed_time, end_time - start_time)
  if i % 50 == 0:
    print('Step {}: {:4.1f}ms'.format(i, (elapsed_time[-50:].mean()) *
1000))
```

```
print('Throughput: {:.0f} images/s'.format(N_run * batch_size /
elapsed_time.sum()))
```

```
Step 0: 1062.6ms
Step 50: 93.0ms
Step 100: 92.8ms
Step 150: 92.1ms
Step 200: 91.5ms
Step 250: 92.4ms
Throughput: 84 images/s
```

TF-TRT FP32 model

We first convert the TF native FP32 model to a TF-TRT FP32 model.

```
print('Converting to TF-TRT FP32...')
conversion_params =
trt.DEFAULT_TRT_CONVERSION_PARAMS._replace(precision_mode=trt.TrtPreci
sionMode.FP32,

max_workspace_size_bytes=8000000000)

converter =
trt.TrtGraphConverterV2(input_saved_model_dir='/content/drive/MyDrive/
Colab Notebooks/inc_saved_model',

conversion_params=conversion_params)
converter.convert()
converter.save(output_saved_model_dir='inc_saved_model_TFTRT_FP32')
print('Done Converting to TF-TRT FP32')
```

```
Converting to TF-TRT FP32...
INFO:tensorflow:Linked TensorRT version: (5, 1, 5)
INFO:tensorflow:Loaded TensorRT version: (5, 1, 5)
INFO:tensorflow:Running against TensorRT version 5.1.5
INFO:tensorflow:Assets written to: inc_saved_model_TFTRT_FP32/assets
Done Converting to TF-TRT FP32
```

```
!saved_model_cli show --all --dir inc_saved_model_TFTRT_FP32
```

```
MetaGraphDef with tag-set: 'serve' contains the following
SignatureDefs:

signature_def['__saved_model_init_op']:
  The given SavedModel SignatureDef contains the following input(s):
  The given SavedModel SignatureDef contains the following output(s):
    outputs['__saved_model_init_op'] tensor_info:
        dtype: DT_INVALID
        shape: unknown_rank
        name: NoOp
```

```
  Method name is:

signature_def['serving_default']:
  The given SavedModel SignatureDef contains the following input(s):
    inputs['input_1'] tensor_info:
        dtype: DT_FLOAT
        shape: (-1, -1, -1, 3)
        name: serving_default_input_1:0
  The given SavedModel SignatureDef contains the following output(s):
    outputs['dense'] tensor_info:
        dtype: DT_FLOAT
        shape: unknown_rank
        name: PartitionedCall:0
  Method name is: tensorflow/serving/predict
```

Next, we load and test the TF-TRT FP32 model.

```
def predict_tftrt(input_saved_model):
    """Runs prediction on a single image and shows the result.
    input_saved_model (string): Name of the input model stored in the
current dir
    """
    img_file = images[0]
    img_path = os.path.join(image_dir,img_file)
    img = image.load_img(img_path, target_size=(224, 224))
    x = image.img_to_array(img)
    x = np.expand_dims(x, axis=0)
    x = preprocess_input(x)
    x = tf.constant(x)

    saved_model_loaded = tf.saved_model.load(input_saved_model,
tags=[tag_constants.SERVING])
    signature_keys = list(saved_model_loaded.signatures.keys())
    print(signature_keys)

    infer = saved_model_loaded.signatures['serving_default']
    print(infer.structured_outputs)

    labeling = infer(x)
    print(labeling.keys())
    #preds = labeling['probs'].numpy()
    preds = labeling['dense'].numpy()
    print(preds.shape)

    p = preds.reshape(-1,1000)
    print(p.shape)

    #print('{} - Predicted: {}'.format(img_path,
decode_predictions(preds, top=3)[0]))
    print('{} - Predicted: {}'.format(img_path, decode_predictions(p,
```

```
top=3)[0]))
    plt.subplot(2,2,1)
    plt.imshow(img);
    plt.axis('off');
    #plt.title(decode_predictions(preds, top=3)[0][0][1])
    plt.title(decode_predictions(p, top=3)[0][0][1])

predict_tftrt('inc_saved_model_TFTRT_FP32')
```

```
['serving_default']
{'dense': TensorSpec(shape=<unknown>, dtype=tf.float32, name='dense')}
dict_keys(['dense'])
(1, 5, 5, 1000)
(25, 1000)
/content/drive/MyDrive/Colab Notebooks/train/Black-grass/0050f38b3.png
- Predicted: [('n07892512', 'red_wine', 0.056537494), ('n03110669',
'cornet', 0.041734096), ('n01729322', 'hognose_snake', 0.032864906)]
```




```
def benchmark_tftrt(input_saved_model):
    saved_model_loaded = tf.saved_model.load(input_saved_model,
tags=[tag_constants.SERVING])
    infer = saved_model_loaded.signatures['serving_default']


    N_run = 300
    elapsed_time = []

    for i in range(N_run):
      start_time = time.time()
      labeling = infer(batched_input)
      #prob = labeling['probs'].numpy()
      end_time = time.time()
      elapsed_time = np.append(elapsed_time, end_time - start_time)
      if i % 50 == 0:
        print('Step {}: {:4.1f}ms'.format(i, (elapsed_time[-
50:].mean()) * 1000))

    print('Throughput: {:.0f} images/s'.format(N_run * batch_size /
elapsed_time.sum()))
```

```
benchmark_tftrt('inc_saved_model_TFTRT_FP32')
```

```
Step 0: 10367.3ms
Step 50: 51.3ms
Step 100: 51.3ms
Step 150: 51.3ms
Step 200: 51.4ms
Step 250: 51.6ms
Throughput: 93 images/s
```

TF-TRT FP16 model

We next convert the native TF FP32 saved model to TF-TRT FP16 model.

```
batch_size = 8
batched_input = np.zeros((batch_size, 224, 224, 3), dtype=np.float32)

for i in range(batch_size):
  img_file = images[i%4]
  img_path = os.path.join(image_dir,img_file)
  img = image.load_img(img_path, target_size=(224, 224))
  x = image.img_to_array(img)
  x = np.expand_dims(x, axis=0)
  x = preprocess_input(x)
  batched_input[i, :] = x
batched_input = tf.constant(batched_input)
print('batched_input shape: ', batched_input.shape)
```

```
batched_input shape:  (8, 224, 224, 3)
```

```
def predict_tftrt(input_saved_model):
    """Runs prediction on a single image and shows the result.
    input_saved_model (string): Name of the input model stored in the
current dir
    """
    img_file = images[0]
    img_path = os.path.join(image_dir,img_file)
    img = image.load_img(img_path, target_size=(224, 224))
    x = image.img_to_array(img)
    x = np.expand_dims(x, axis=0)
    x = preprocess_input(x)
    x = tf.constant(x)

    saved_model_loaded = tf.saved_model.load(input_saved_model,
tags=[tag_constants.SERVING])
    signature_keys = list(saved_model_loaded.signatures.keys())
    print(signature_keys)

    infer = saved_model_loaded.signatures['serving_default']
    print(infer.structured_outputs)
```

```
    labeling = infer(x)

    preds = labeling['dense'].numpy()
    print(preds.shape)

    p = preds.reshape(-1,1000)
    print(p.shape)


    print('{} - Predicted: {}'.format(img_path, decode_predictions(p,
top=3)[0]))
    plt.subplot(2,2,1)
    plt.imshow(img);
    plt.axis('off');
    plt.title(decode_predictions(p, top=3)[0][0][1])

print('Converting to TF-TRT FP16...')
conversion_params = trt.DEFAULT_TRT_CONVERSION_PARAMS._replace(
    precision_mode=trt.TrtPrecisionMode.FP16,
    max_workspace_size_bytes=8000000000)
converter = trt.TrtGraphConverterV2(
   input_saved_model_dir='/content/drive/MyDrive/Colab
Notebooks/inc_saved_model', conversion_params=conversion_params)
converter.convert()
converter.save(output_saved_model_dir='inc_saved_model_TFTRT_FP16')
print('Done Converting to TF-TRT FP16')
```

```
Converting to TF-TRT FP16...
INFO:tensorflow:Linked TensorRT version: (5, 1, 5)
INFO:tensorflow:Loaded TensorRT version: (5, 1, 5)
INFO:tensorflow:Running against TensorRT version 5.1.5
INFO:tensorflow:Assets written to: inc_saved_model_TFTRT_FP16/assets
Done Converting to TF-TRT FP16
```

```
predict_tftrt('inc_saved_model_TFTRT_FP16')
```

```
['serving_default']
{'dense': TensorSpec(shape=<unknown>, dtype=tf.float32, name='dense')}
(1, 5, 5, 1000)
(25, 1000)
/content/drive/MyDrive/Colab Notebooks/train/Black-grass/0050f38b3.png
- Predicted: [('n04266014', 'space_shuttle', 0.03454123),
('n02091244', 'Ibizan_hound', 0.032830097), ('n02106166',
'Border_collie', 0.028929843)]
```




```
def benchmark_tftrt(input_saved_model):
    saved_model_loaded = tf.saved_model.load(input_saved_model,
tags=[tag_constants.SERVING])
    infer = saved_model_loaded.signatures['serving_default']


    N_run = 300
    elapsed_time = []

    for i in range(N_run):
      start_time = time.time()
      labeling = infer(batched_input)
      #prob = labeling['probs'].numpy()
      end_time = time.time()
      elapsed_time = np.append(elapsed_time, end_time - start_time)
      if i % 50 == 0:
        print('Step {}: {:4.1f}ms'.format(i, (elapsed_time[-
50:].mean()) * 1000))

    print('Throughput: {:.0f} images/s'.format(N_run * batch_size /
elapsed_time.sum()))

benchmark_tftrt('inc_saved_model_TFTRT_FP16')
```

```
Step 0: 24412.4ms
Step 50: 49.5ms
Step 100: 49.8ms
Step 150: 49.6ms
Step 200: 49.7ms
Step 250: 49.7ms
Throughput: 61 images/s
```

TF-TRT INT8 model

Creating TF-TRT INT8 model requires a small calibration dataset. This data set ideally should represent the test data in production well, and will be used to create a value histogram for each layer in the neural network for effective 8-bit quantization.

Herein, for demonstration purposes, we take only the 4 images that we downloaded for calibration. In production, this set should be more representative of the production data.

Due to Colab memory limit which sometime causes TensorRT to crash, to proceed, first, restart the runtime by pressing CTRL+M or select *Runtime -> Restart runtime...* or simply execute the next cell.

```
# import os
# os.kill(os.getpid(), 9)
```

```
batch_size = 8
batched_input = np.zeros((batch_size, 224, 224, 3), dtype=np.float32)

for i in range(batch_size):
  img_file = images[i%4]
  img_path = os.path.join(image_dir,img_file)
  img = image.load_img(img_path, target_size=(224, 224))
  x = image.img_to_array(img)
  x = np.expand_dims(x, axis=0)
  x = preprocess_input(x)
  batched_input[i, :] = x
batched_input = tf.constant(batched_input)
print('batched_input shape: ', batched_input.shape)
```

```
batched_input shape:  (8, 224, 224, 3)
```

```
print('Converting to TF-TRT INT8...')
conversion_params = trt.DEFAULT_TRT_CONVERSION_PARAMS._replace(
    precision_mode=trt.TrtPrecisionMode.INT8,
    max_workspace_size_bytes=8000000000,
    use_calibration=True)
converter = trt.TrtGraphConverterV2(
    input_saved_model_dir='/content/drive/MyDrive/Colab
Notebooks/inc_saved_model',
    conversion_params=conversion_params)

def calibration_input_fn():
    yield (batched_input, )
converter.convert(calibration_input_fn=calibration_input_fn)

converter.save(output_saved_model_dir='inc_saved_model_TFTRT_INT8')
print('Done Converting to TF-TRT INT8')
```

```
Converting to TF-TRT INT8...
INFO:tensorflow:Linked TensorRT version: (5, 1, 5)
INFO:tensorflow:Loaded TensorRT version: (5, 1, 5)
INFO:tensorflow:Running against TensorRT version 5.1.5
WARNING:tensorflow:From
/usr/local/lib/python3.7/dist-packages/tensorflow_core/python/ops/
resource_variable_ops.py:1781: calling BaseResourceVariable.__init__
(from tensorflow.python.ops.resource_variable_ops) with constraint is
deprecated and will be removed in a future version.
Instructions for updating:
If using Keras pass *_constraint arguments to layers.
```

```
INFO:tensorflow:Assets written to: inc_saved_model_TFTRT_INT8/assets
Done Converting to TF-TRT INT8
```

```
def predict_tftrt(input_saved_model):
    """Runs prediction on a single image and shows the result.
    input_saved_model (string): Name of the input model stored in the
current dir
    """
    img_file = images[0]
    img_path = os.path.join(image_dir,img_file)
    img = image.load_img(img_path, target_size=(224, 224))
    x = image.img_to_array(img)
    x = np.expand_dims(x, axis=0)
    x = preprocess_input(x)
    x = tf.constant(x)

    saved_model_loaded = tf.saved_model.load(input_saved_model,
tags=[tag_constants.SERVING])
    signature_keys = list(saved_model_loaded.signatures.keys())
    print(signature_keys)

    infer = saved_model_loaded.signatures['serving_default']
    print(infer.structured_outputs)

    labeling = infer(x)
    preds = labeling['dense'].numpy()
    p = preds.reshape(-1,1000)
    print('{} - Predicted: {}'.format(img_path, decode_predictions(p,
top=3)[0]))
    plt.subplot(2,2,1)
    plt.imshow(img);
    plt.axis('off');
    plt.title(decode_predictions(p, top=3)[0][0][1])

predict_tftrt('inc_saved_model_TFTRT_INT8')
```

```
WARNING:tensorflow:Unresolved object in checkpoint:
(root).trt_engine_resources.TRTEngineOp_0._serialized_trt_resource_fil
ename
WARNING:tensorflow:Unresolved object in checkpoint:
(root).trt_engine_resources.TRTEngineOp_1._serialized_trt_resource_fil
ename
WARNING:tensorflow:A checkpoint was restored (e.g.
tf.train.Checkpoint.restore or tf.keras.Model.load_weights) but not
all checkpointed values were used. See above for specific issues. Use
expect_partial() on the load status object, e.g.
tf.train.Checkpoint.restore(...).expect_partial(), to silence these
warnings, or use assert_consumed() to make the check explicit. See
https://www.tensorflow.org/alpha/guide/checkpoints#loading_mechanics
for details.
['serving_default']
```

```
{'dense': TensorSpec(shape=<unknown>, dtype=tf.float32, name='dense')}
/content/drive/MyDrive/Colab Notebooks/train/Black-grass/0050f38b3.png
- Predicted: [('n04266014', 'space_shuttle', 0.03462654),
('n02091244', 'Ibizan_hound', 0.032453127), ('n02106166',
'Border_collie', 0.029094027)]
```




```
def benchmark_tftrt(input_saved_model):
    saved_model_loaded = tf.saved_model.load(input_saved_model,
tags=[tag_constants.SERVING])
    infer = saved_model_loaded.signatures['serving_default']

    N_run = 300
    elapsed_time = []

    for i in range(N_run):
      start_time = time.time()
      labeling = infer(batched_input)
      #prob = labeling['probs'].numpy()
      end_time = time.time()
      elapsed_time = np.append(elapsed_time, end_time - start_time)
      if i % 50 == 0:
        print('Step {}: {:4.1f}ms'.format(i, (elapsed_time[-
50:].mean()) * 1000))

    print('Throughput: {:.0f} images/s'.format(N_run * batch_size /
elapsed_time.sum()))

benchmark_tftrt('inc_saved_model_TFTRT_INT8')
```

```
WARNING:tensorflow:Unresolved object in checkpoint:
(root).trt_engine_resources.TRTEngineOp_0._serialized_trt_resource_fil
ename
WARNING:tensorflow:Unresolved object in checkpoint:
(root).trt_engine_resources.TRTEngineOp_1._serialized_trt_resource_fil
ename
WARNING:tensorflow:A checkpoint was restored (e.g.
tf.train.Checkpoint.restore or tf.keras.Model.load_weights) but not
all checkpointed values were used. See above for specific issues. Use
expect_partial() on the load status object, e.g.
tf.train.Checkpoint.restore(...).expect_partial(), to silence these
```

```
warnings, or use assert_consumed() to make the check explicit. See
https://www.tensorflow.org/alpha/guide/checkpoints#loading_mechanics
for details.
Step 0: 3354.7ms
Step 50: 51.5ms
Step 100: 49.6ms
Step 150: 49.7ms
Step 200: 49.6ms
Step 250: 49.5ms
Throughput: 131 images/s
```

Conclusion

In this notebook, we have demonstrated the process of creating TF-TRT FP32, FP16 and INT8 inference models from an original Keras FP32 model, as well as verify their speed and accuracy.

What's next

Try TF-TRT on your own model and data, and experience the simplicity and speed up it offers.

```
!pip install pillow matplotlib
!pip install tensorflow-gpu==2.0.0
```

```
Requirement already satisfied: pillow in
/usr/local/lib/python3.7/dist-packages (7.1.2)
Requirement already satisfied: matplotlib in
/usr/local/lib/python3.7/dist-packages (3.2.2)
Requirement already satisfied: cycler>=0.10 in
/usr/local/lib/python3.7/dist-packages (from matplotlib) (0.11.0)
Requirement already satisfied: pyparsing!=2.0.4,!=2.1.2,!
=2.1.6,>=2.0.1 in /usr/local/lib/python3.7/dist-packages (from
matplotlib) (3.0.7)
Requirement already satisfied: python-dateutil>=2.1 in
/usr/local/lib/python3.7/dist-packages (from matplotlib) (2.8.2)
Requirement already satisfied: kiwisolver>=1.0.1 in
/usr/local/lib/python3.7/dist-packages (from matplotlib) (1.4.0)
Requirement already satisfied: numpy>=1.11 in
/usr/local/lib/python3.7/dist-packages (from matplotlib) (1.21.5)
Requirement already satisfied: typing-extensions in
/usr/local/lib/python3.7/dist-packages (from kiwisolver>=1.0.1-
>matplotlib) (3.10.0.2)
Requirement already satisfied: six>=1.5 in
/usr/local/lib/python3.7/dist-packages (from python-dateutil>=2.1-
>matplotlib) (1.15.0)
Requirement already satisfied: tensorflow-gpu==2.0.0 in
/usr/local/lib/python3.7/dist-packages (2.0.0)
Requirement already satisfied: google-pasta>=0.1.6 in
/usr/local/lib/python3.7/dist-packages (from tensorflow-gpu==2.0.0)
(0.2.0)
Requirement already satisfied: numpy<2.0,>=1.16.0 in
/usr/local/lib/python3.7/dist-packages (from tensorflow-gpu==2.0.0)
(1.21.5)
Requirement already satisfied: gast==0.2.2 in
/usr/local/lib/python3.7/dist-packages (from tensorflow-gpu==2.0.0)
(0.2.2)
Requirement already satisfied: astor>=0.6.0 in
/usr/local/lib/python3.7/dist-packages (from tensorflow-gpu==2.0.0)
(0.8.1)
Requirement already satisfied: keras-preprocessing>=1.0.5 in
/usr/local/lib/python3.7/dist-packages (from tensorflow-gpu==2.0.0)
(1.1.2)
Requirement already satisfied: termcolor>=1.1.0 in
/usr/local/lib/python3.7/dist-packages (from tensorflow-gpu==2.0.0)
(1.1.0)
Requirement already satisfied: protobuf>=3.6.1 in
/usr/local/lib/python3.7/dist-packages (from tensorflow-gpu==2.0.0)
(3.17.3)
Requirement already satisfied: opt-einsum>=2.3.2 in
/usr/local/lib/python3.7/dist-packages (from tensorflow-gpu==2.0.0)
(3.3.0)
```

```
Requirement already satisfied: absl-py>=0.7.0 in
/usr/local/lib/python3.7/dist-packages (from tensorflow-gpu==2.0.0)
(1.0.0)
Requirement already satisfied: tensorflow-estimator<2.1.0,>=2.0.0
in /usr/local/lib/python3.7/dist-packages (from tensorflow-gpu==2.0.0)
(2.0.1)
Requirement already satisfied: grpcio>=1.8.6 in
/usr/local/lib/python3.7/dist-packages (from tensorflow-gpu==2.0.0)
(1.44.0)
Requirement already satisfied: six>=1.10.0 in
/usr/local/lib/python3.7/dist-packages (from tensorflow-gpu==2.0.0)
(1.15.0)
Requirement already satisfied: wheel>=0.26 in
/usr/local/lib/python3.7/dist-packages (from tensorflow-gpu==2.0.0)
(0.37.1)
Requirement already satisfied: keras-applications>=1.0.8 in
/usr/local/lib/python3.7/dist-packages (from tensorflow-gpu==2.0.0)
(1.0.8)
Requirement already satisfied: tensorboard<2.1.0,>=2.0.0 in
/usr/local/lib/python3.7/dist-packages (from tensorflow-gpu==2.0.0)
(2.0.2)
Requirement already satisfied: wrapt>=1.11.1 in
/usr/local/lib/python3.7/dist-packages (from tensorflow-gpu==2.0.0)
(1.14.0)
Requirement already satisfied: h5py in /usr/local/lib/python3.7/dist-
packages (from keras-applications>=1.0.8->tensorflow-gpu==2.0.0)
(3.1.0)
Requirement already satisfied: google-auth-oauthlib<0.5,>=0.4.1 in
/usr/local/lib/python3.7/dist-packages (from
tensorboard<2.1.0,>=2.0.0->tensorflow-gpu==2.0.0) (0.4.6)
Requirement already satisfied: markdown>=2.6.8 in
/usr/local/lib/python3.7/dist-packages (from
tensorboard<2.1.0,>=2.0.0->tensorflow-gpu==2.0.0) (3.3.6)
Requirement already satisfied: setuptools>=41.0.0 in
/usr/local/lib/python3.7/dist-packages (from
tensorboard<2.1.0,>=2.0.0->tensorflow-gpu==2.0.0) (57.4.0)
Requirement already satisfied: requests<3,>=2.21.0 in
/usr/local/lib/python3.7/dist-packages (from
tensorboard<2.1.0,>=2.0.0->tensorflow-gpu==2.0.0) (2.23.0)
Requirement already satisfied: google-auth<2,>=1.6.3 in
/usr/local/lib/python3.7/dist-packages (from
tensorboard<2.1.0,>=2.0.0->tensorflow-gpu==2.0.0) (1.35.0)
Requirement already satisfied: werkzeug>=0.11.15 in
/usr/local/lib/python3.7/dist-packages (from
tensorboard<2.1.0,>=2.0.0->tensorflow-gpu==2.0.0) (1.0.1)
Requirement already satisfied: rsa<5,>=3.1.4 in
/usr/local/lib/python3.7/dist-packages (from google-auth<2,>=1.6.3-
>tensorboard<2.1.0,>=2.0.0->tensorflow-gpu==2.0.0) (4.8)
Requirement already satisfied: cachetools<5.0,>=2.0.0 in
/usr/local/lib/python3.7/dist-packages (from google-auth<2,>=1.6.3-
```

```
>tensorboard<2.1.0,>=2.0.0->tensorflow-gpu==2.0.0) (4.2.4)
Requirement already satisfied: pyasn1-modules>=0.2.1 in
/usr/local/lib/python3.7/dist-packages (from google-auth<2,>=1.6.3-
>tensorboard<2.1.0,>=2.0.0->tensorflow-gpu==2.0.0) (0.2.8)
Requirement already satisfied: requests-oauthlib>=0.7.0 in
/usr/local/lib/python3.7/dist-packages (from google-auth-
oauthlib<0.5,>=0.4.1->tensorboard<2.1.0,>=2.0.0->tensorflow-
gpu==2.0.0) (1.3.1)
Requirement already satisfied: importlib-metadata>=4.4 in
/usr/local/lib/python3.7/dist-packages (from markdown>=2.6.8-
>tensorboard<2.1.0,>=2.0.0->tensorflow-gpu==2.0.0) (4.11.3)
Requirement already satisfied: zipp>=0.5 in
/usr/local/lib/python3.7/dist-packages (from importlib-metadata>=4.4-
>markdown>=2.6.8->tensorboard<2.1.0,>=2.0.0->tensorflow-gpu==2.0.0)
(3.7.0)
Requirement already satisfied: typing-extensions>=3.6.4 in
/usr/local/lib/python3.7/dist-packages (from importlib-metadata>=4.4-
>markdown>=2.6.8->tensorboard<2.1.0,>=2.0.0->tensorflow-gpu==2.0.0)
(3.10.0.2)
Requirement already satisfied: pyasn1<0.5.0,>=0.4.6 in
/usr/local/lib/python3.7/dist-packages (from pyasn1-modules>=0.2.1-
>google-auth<2,>=1.6.3->tensorboard<2.1.0,>=2.0.0->tensorflow-
gpu==2.0.0) (0.4.8)
Requirement already satisfied: idna<3,>=2.5 in
/usr/local/lib/python3.7/dist-packages (from requests<3,>=2.21.0-
>tensorboard<2.1.0,>=2.0.0->tensorflow-gpu==2.0.0) (2.10)
Requirement already satisfied: certifi>=2017.4.17 in
/usr/local/lib/python3.7/dist-packages (from requests<3,>=2.21.0-
>tensorboard<2.1.0,>=2.0.0->tensorflow-gpu==2.0.0) (2021.10.8)
Requirement already satisfied: chardet<4,>=3.0.2 in
/usr/local/lib/python3.7/dist-packages (from requests<3,>=2.21.0-
>tensorboard<2.1.0,>=2.0.0->tensorflow-gpu==2.0.0) (3.0.4)
Requirement already satisfied: urllib3!=1.25.0,!=1.25.1,<1.26,>=1.21.1
in /usr/local/lib/python3.7/dist-packages (from requests<3,>=2.21.0-
>tensorboard<2.1.0,>=2.0.0->tensorflow-gpu==2.0.0) (1.24.3)
Requirement already satisfied: oauthlib>=3.0.0 in
/usr/local/lib/python3.7/dist-packages (from requests-oauthlib>=0.7.0-
>google-auth-oauthlib<0.5,>=0.4.1->tensorboard<2.1.0,>=2.0.0-
>tensorflow-gpu==2.0.0) (3.2.0)
Requirement already satisfied: cached-property in
/usr/local/lib/python3.7/dist-packages (from h5py->keras-
applications>=1.0.8->tensorflow-gpu==2.0.0) (1.5.2)
```

```
!pip install 'h5py==2.10.0'
```

```
Collecting h5py==2.10.0
  Downloading h5py-2.10.0-cp37-cp37m-manylinux1_x86_64.whl (2.9 MB)
ent already satisfied: six in /usr/local/lib/python3.7/dist-packages
(from h5py==2.10.0) (1.15.0)
Requirement already satisfied: numpy>=1.7 in
/usr/local/lib/python3.7/dist-packages (from h5py==2.10.0) (1.21.5)
```

```
Installing collected packages: h5py
  Attempting uninstall: h5py
    Found existing installation: h5py 3.1.0
    Uninstalling h5py-3.1.0:
      Successfully uninstalled h5py-3.1.0
ERROR: pip's dependency resolver does not currently take into account
all the packages that are installed. This behaviour is the source of
the following dependency conflicts.
tensorflow 2.8.0 requires tf-estimator-nightly==2.8.0.dev2021122109,
which is not installed.
tensorflow 2.8.0 requires tensorboard<2.9,>=2.8, but you have
tensorboard 2.0.2 which is incompatible.
Successfully installed h5py-2.10.0
```

```
from google.colab import drive
drive.mount('/content/drive')
```

```
Drive already mounted at /content/drive; to attempt to forcibly
remount, call drive.mount("/content/drive", force_remount=True).
```

```
# Copyright 2019 NVIDIA Corporation. All Rights Reserved.
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#
==========================================================================
========
```

TF-TRT Inference from Keras Model with TensorFlow 2.0

Introduction

The NVIDIA TensorRT is a C++ library that facilitates high performance inference on NVIDIA graphics processing units (GPUs). TensorRT takes a trained network, which consists of a network definition and a set of trained parameters, and produces a highly optimized runtime engine which performs inference for that network.

TensorFlow™ integration with TensorRT™ (TF-TRT) optimizes and executes compatible subgraphs, allowing TensorFlow to execute the remaining graph. While you can still use TensorFlow's wide and flexible feature set, TensorRT will parse the model and apply optimizations to the portions of the graph wherever possible.

In this notebook, we demonstrate the process of creating a TF-TRT optimized model from a VGG-16 Keras saved model.

Requirement

GPU

Before running this notebook, please set the Colab runtime environment to GPU via the menu *Runtime => Change runtime type => GPU*.

This demo will work on any NVIDIA GPU with CUDA cores, though for improved FP16 and INT8 inference, a Volta, Turing or newer generation GPU with Tensor cores is desired. On Google Colab, this normally means a T4 GPU. If you are assigned an older K80 GPU, another trial at another time might give you a T4 GPU.

```
!nvidia-smi
```

```
Sat Apr  9 06:25:28 2022
+-----------------------------------------------------------------------------
--------+
| NVIDIA-SMI 460.32.03    Driver Version: 460.32.03    CUDA Version:
11.2     |
|-------------------------------+----------------------
+----------------------+
| GPU  Name        Persistence-M| Bus-Id        Disp.A | Volatile
Uncorr. ECC |
| Fan  Temp  Perf  Pwr:Usage/Cap|         Memory-Usage | GPU-Util
Compute M. |
|                               |                      |
MIG M. |
|
===============================+======================+===============
=======|
|   0  Tesla K80           Off  | 00000000:00:04.0 Off |
0 |
| N/A   64C    P8    32W / 149W |      0MiB / 11441MiB |      0%
Default |
|                               |                      |
N/A |
+-------------------------------+----------------------
+----------------------+


+-----------------------------------------------------------------------------
--------+
```

```
| Processes:
|
|  GPU   GI   CI        PID   Type   Process name                  GPU
Memory |
|        ID   ID
Usage      |
|
=============================================================================
=======|
|  No running processes found
|
+-----------------------------------------------------------------------------
--------+
```

Install TensorFlow-GPU 2.0

```
import tensorflow as tf
print("Tensorflow version: ", tf.version.VERSION)
```

```
Tensorflow version:  2.0.0
```

Install TensorRT Runtime

```
%%bash
wget https://developer.download.nvidia.com/compute/machine-
learning/repos/ubuntu1804/x86_64/nvidia-machine-learning-repo-
ubuntu1804_1.0.0-1_amd64.deb

dpkg -i nvidia-machine-learning-repo-*.deb
apt-get update

sudo apt-get install libnvinfer5
```

```
Selecting previously unselected package nvidia-machine-learning-repo-
ubuntu1804.
(Reading database ... 156210 files and directories currently
installed.)
Preparing to unpack nvidia-machine-learning-repo-ubuntu1804_1.0.0-
1_amd64.deb ...
Unpacking nvidia-machine-learning-repo-ubuntu1804 (1.0.0-1) ...
Setting up nvidia-machine-learning-repo-ubuntu1804 (1.0.0-1) ...
Get:1 https://cloud.r-project.org/bin/linux/ubuntu bionic-cran40/
InRelease [3,626 B]
Ign:2
http://developer.download.nvidia.com/compute/machine-learning/repos/
ubuntu1804/x86_64  InRelease
Hit:3
http://developer.download.nvidia.com/compute/machine-learning/repos/
ubuntu1804/x86_64  Release
Ign:4
https://developer.download.nvidia.com/compute/cuda/repos/ubuntu1804/
x86_64  InRelease
Get:5
```

```
https://developer.download.nvidia.com/compute/cuda/repos/ubuntu1804/
x86_64  Release [696 B]
Get:6
https://developer.download.nvidia.com/compute/cuda/repos/ubuntu1804/
x86_64  Release.gpg [836 B]
Get:7 http://security.ubuntu.com/ubuntu bionic-security InRelease
[88.7 kB]
Get:8 http://ppa.launchpad.net/c2d4u.team/c2d4u4.0+/ubuntu bionic
InRelease [15.9 kB]
Hit:9 http://archive.ubuntu.com/ubuntu bionic InRelease
Get:10 https://cloud.r-project.org/bin/linux/ubuntu bionic-cran40/
Packages [82.3 kB]
Get:11 http://archive.ubuntu.com/ubuntu bionic-updates InRelease [88.7
kB]
Hit:13 http://ppa.launchpad.net/cran/libgit2/ubuntu bionic InRelease
Get:14 http://archive.ubuntu.com/ubuntu bionic-backports InRelease
[74.6 kB]
Get:15
https://developer.download.nvidia.com/compute/cuda/repos/ubuntu1804/
x86_64  Packages [952 kB]
Get:16 http://ppa.launchpad.net/deadsnakes/ppa/ubuntu bionic InRelease
[15.9 kB]
Get:17 http://ppa.launchpad.net/graphics-drivers/ppa/ubuntu bionic
InRelease [21.3 kB]
Get:18 http://security.ubuntu.com/ubuntu bionic-security/universe
amd64 Packages [1,490 kB]
Get:19 http://ppa.launchpad.net/c2d4u.team/c2d4u4.0+/ubuntu
bionic/main Sources [1,831 kB]
Get:20 http://archive.ubuntu.com/ubuntu bionic-updates/universe amd64
Packages [2,268 kB]
Get:21 http://security.ubuntu.com/ubuntu bionic-security/restricted
amd64 Packages [884 kB]
Get:22 http://security.ubuntu.com/ubuntu bionic-security/main amd64
Packages [2,694 kB]
Get:23 http://archive.ubuntu.com/ubuntu bionic-updates/multiverse
amd64 Packages [29.8 kB]
Get:24 http://archive.ubuntu.com/ubuntu bionic-updates/restricted
amd64 Packages [918 kB]
Get:25 http://archive.ubuntu.com/ubuntu bionic-updates/main amd64
Packages [3,133 kB]
Get:26 http://ppa.launchpad.net/c2d4u.team/c2d4u4.0+/ubuntu
bionic/main amd64 Packages [938 kB]
Get:27 http://archive.ubuntu.com/ubuntu bionic-backports/main amd64
Packages [12.2 kB]
Get:28 http://archive.ubuntu.com/ubuntu bionic-backports/universe
amd64 Packages [12.9 kB]
Get:29 http://ppa.launchpad.net/deadsnakes/ppa/ubuntu bionic/main
amd64 Packages [45.2 kB]
Get:30 http://ppa.launchpad.net/graphics-drivers/ppa/ubuntu
bionic/main amd64 Packages [44.3 kB]
```

```
Fetched 15.6 MB in 4s (3,941 kB/s)
Reading package lists...
Reading package lists...
Building dependency tree...
Reading state information...
The following NEW packages will be installed:
  libnvinfer5
0 upgraded, 1 newly installed, 0 to remove and 96 not upgraded.
Need to get 46.7 MB of archives.
After this operation, 163 MB of additional disk space will be used.
Get:1
http://developer.download.nvidia.com/compute/machine-learning/repos/
ubuntu1804/x86_64  libnvinfer5 5.1.5-1+cuda10.1 [46.7 MB]
Fetched 46.7 MB in 1s (53.6 MB/s)
Selecting previously unselected package libnvinfer5.
(Reading database ...  (Reading database ... 5% (Reading database ...
10% (Reading database ... 15% (Reading database ... 20% (Reading
database ... 25% (Reading database ... 30% (Reading database ... 35%
(Reading database ... 40% (Reading database ... 45% (Reading
database ... 50% (Reading database ... 55% (Reading database ... 60%
(Reading database ... 65% (Reading database ... 70% (Reading
database ... 75% (Reading database ... 80% (Reading database ... 85%
(Reading database ... 90% (Reading database ... 95% (Reading
database ... 100% (Reading database ... 156213 files and directories
currently installed.)
Preparing to unpack .../libnvinfer5_5.1.5-1+cuda10.1_amd64.deb ...
Unpacking libnvinfer5 (5.1.5-1+cuda10.1) ...
Setting up libnvinfer5 (5.1.5-1+cuda10.1) ...
Processing triggers for libc-bin (2.27-3ubuntu1.3) ...
/sbin/ldconfig.real:
/usr/local/lib/python3.7/dist-packages/ideep4py/lib/libmkldnn.so.0 is
not a symbolic link


--2022-04-09 06:25:47--
https://developer.download.nvidia.com/compute/machine-learning/repos/
ubuntu1804/x86_64/nvidia-machine-learning-repo-ubuntu1804_1.0.0-
1_amd64.deb
Resolving developer.download.nvidia.com
(developer.download.nvidia.com)... 152.195.19.142
Connecting to developer.download.nvidia.com
(developer.download.nvidia.com)|152.195.19.142|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 2926 (2.9K) [application/x-deb]
Saving to: ‘nvidia-machine-learning-repo-ubuntu1804_1.0.0-1_amd64.deb’

     0K ..                                                    100%
109M=0s

2022-04-09 06:25:47 (109 MB/s) - ‘nvidia-machine-learning-repo-
```

```
ubuntu1804_1.0.0-1_amd64.deb’ saved [2926/2926]

W: Target Packages (Packages) is configured multiple times in
/etc/apt/sources.list.d/nvidia-machine-learning.list:1 and
/etc/apt/sources.list.d/nvidia-ml.list:1
W: Target Packages (Packages) is configured multiple times in
/etc/apt/sources.list.d/nvidia-machine-learning.list:1 and
/etc/apt/sources.list.d/nvidia-ml.list:1
debconf: unable to initialize frontend: Dialog
debconf: (No usable dialog-like program is installed, so the dialog
based frontend cannot be used. at
/usr/share/perl5/Debconf/FrontEnd/Dialog.pm line 76, <> line 1.)
debconf: falling back to frontend: Readline
debconf: unable to initialize frontend: Readline
debconf: (This frontend requires a controlling tty.)
debconf: falling back to frontend: Teletype
dpkg-preconfigure: unable to re-open stdin:
W: Target Packages (Packages) is configured multiple times in
/etc/apt/sources.list.d/nvidia-machine-learning.list:1 and
/etc/apt/sources.list.d/nvidia-ml.list:1
```

```
# check TensorRT version
print("TensorRT version: ")
!dpkg -l | grep nvinfer
```

```
TensorRT version:
ii  libnvinfer5                              5.1.5-1+cuda10.1
amd64        TensorRT runtime libraries
```

A successfull TensorRT installation looks like:

```
TensorRT version:
ii  libnvinfer5   5.1.5-1+cuda10.1   amd64        TensorRT runtime
libraries
```

Check Tensor core GPU

The below code check whether a Tensor-core GPU is present.

```
from tensorflow.python.client import device_lib

def check_tensor_core_gpu_present():
    local_device_protos = device_lib.list_local_devices()
    for line in local_device_protos:
        if "compute capability" in str(line):
            compute_capability =
float(line.physical_device_desc.split("compute capability: ")[-1])
            if compute_capability>=7.0:
                return True
```

```
print("Tensor Core GPU Present:", check_tensor_core_gpu_present())
tensor_core_gpu = check_tensor_core_gpu_present()
```

```
Tensor Core GPU Present: None
```

Importing required libraries

```
from __future__ import absolute_import, division, print_function,
unicode_literals
import os
import time

import numpy as np
import matplotlib.pyplot as plt

import tensorflow as tf
from tensorflow import keras
from tensorflow.python.compiler.tensorrt import trt_convert as trt
from tensorflow.python.saved_model import tag_constants
from tensorflow.keras.applications.resnet50 import ResNet50
from tensorflow.keras.preprocessing import image
from tensorflow.keras.applications.resnet50 import preprocess_input,
decode_predictions
from tensorflow.keras.applications.vgg16 import
VGG16,preprocess_input, decode_predictions

from tensorflow.keras.applications.inception_v3 import
InceptionV3,preprocess_input, decode_predictions
```

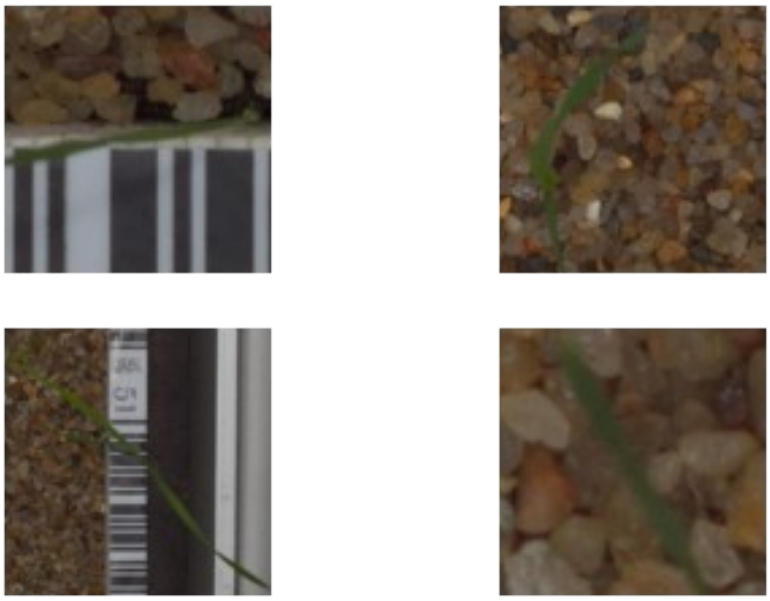
```
image_dir = '/content/drive/MyDrive/Colab Notebooks/test'
images = os.listdir(image_dir)
images[1]
```

```
{"type":"string"}
```

```
from tensorflow.keras.preprocessing import image

fig, axes = plt.subplots(nrows=2, ncols=2)

for i in range(4):
  img_file = images[i]
  img_path = os.path.join(image_dir,img_file)
  img = image.load_img(img_path, target_size=(224, 224))
  plt.subplot(2,2,i+1)
  plt.imshow(img);
  plt.axis('off');
```



Model

We next download and test a VGG16 pre-trained model from the Keras model zoo.

```
# model = ResNet50(weights='imagenet')
#model = InceptionV3(weights = 'imagenet')
model = VGG16(weights= 'imagenet')
```

```
Downloading data from https://github.com/fchollet/deep-learning-
models/releases/download/v0.1/
vgg16_weights_tf_dim_ordering_tf_kernels.h5
553467904/553467096 [==============================] - 5s 0us/step
```

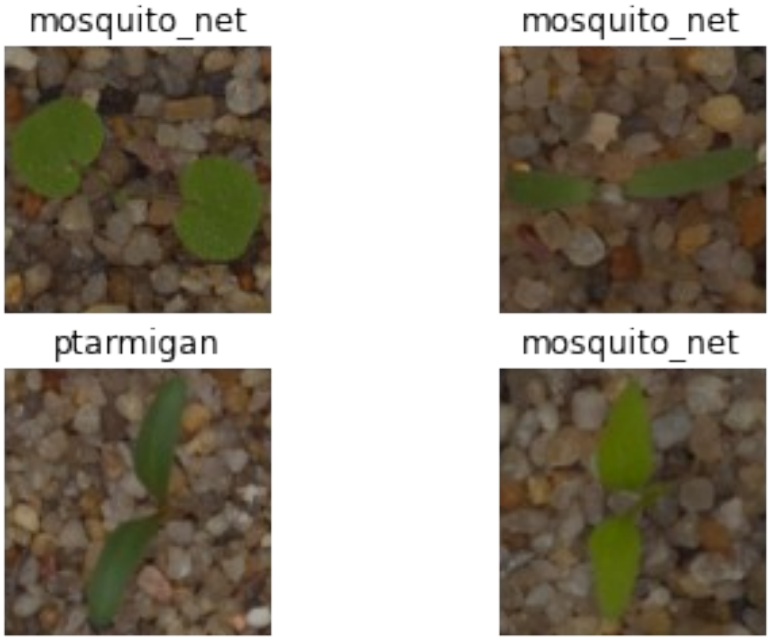
```
for i in range(4):
  img_file = images[i]
  img_path = os.path.join(image_dir,img_file)
  img = image.load_img(img_path, target_size=(224, 224))
  x = image.img_to_array(img)
  x = np.expand_dims(x, axis=0)
  x = preprocess_input(x)

  preds = model.predict(x)
  # decode the results into a list of tuples (class, description,
probability)
  # (one such list for each sample in the batch)
  print('{} - Predicted: {}'.format(img_path,
decode_predictions(preds, top=3)[0]))

  plt.subplot(2,2,i+1)
  plt.imshow(img);
  plt.axis('off');
```

```
  plt.title(decode_predictions(preds, top=3)[0][0][1])
```

```
Downloading data from
https://storage.googleapis.com/download.tensorflow.org/data/imagenet_c
lass_index.json
40960/35363 [==================================] - 0s 0us/step
/content/drive/MyDrive/Colab Notebooks/test/0021e90e4.png - Predicted:
[('n03788365', 'mosquito_net', 0.04051131), ('n07860988', 'dough',
0.016027138), ('n01498041', 'stingray', 0.014224683)]
/content/drive/MyDrive/Colab Notebooks/test/003d61042.png - Predicted:
[('n03788365', 'mosquito_net', 0.037750714), ('n15075141',
'toilet_tissue', 0.014821087), ('n07860988', 'dough', 0.014235996)]
/content/drive/MyDrive/Colab Notebooks/test/007b3da8b.png - Predicted:
[('n01796340', 'ptarmigan', 0.015751045), ('n03788365',
'mosquito_net', 0.014894152), ('n07860988', 'dough', 0.0144516155)]
/content/drive/MyDrive/Colab Notebooks/test/0086a6340.png - Predicted:
[('n03788365', 'mosquito_net', 0.021375598), ('n01796340',
'ptarmigan', 0.016553488), ('n07860988', 'dough', 0.012831482)]
```



TF-TRT takes input as aTensorFlow saved model, therefore, we re-export the Keras model as a TF saved model.

```
# Save the entire model as a SavedModel.
model.save('/content/drive/MyDrive/Colab Notebooks/vgg16_saved_model')
```

```
WARNING:tensorflow:From
/usr/local/lib/python3.7/dist-packages/tensorflow_core/python/ops/
resource_variable_ops.py:1781: calling BaseResourceVariable.__init__
(from tensorflow.python.ops.resource_variable_ops) with constraint is
deprecated and will be removed in a future version.
```

```
Instructions for updating:
If using Keras pass *_constraint arguments to layers.
INFO:tensorflow:Assets written to: /content/drive/MyDrive/Colab
Notebooks/vgg16_saved_model/assets
```

```
!saved_model_cli show --all --dir vgg16_saved_model
```

```
Traceback (most recent call last):
  File "/usr/local/bin/saved_model_cli", line 8, in <module>
    sys.exit(main())
  File
"/usr/local/lib/python3.7/dist-packages/tensorflow_core/python/tools/
saved_model_cli.py", line 909, in main
    args.func(args)
  File
"/usr/local/lib/python3.7/dist-packages/tensorflow_core/python/tools/
saved_model_cli.py", line 610, in show
    _show_all(args.dir)
  File
"/usr/local/lib/python3.7/dist-packages/tensorflow_core/python/tools/
saved_model_cli.py", line 192, in _show_all
    tag_sets =
saved_model_utils.get_saved_model_tag_sets(saved_model_dir)
  File
"/usr/local/lib/python3.7/dist-packages/tensorflow_core/python/tools/
saved_model_utils.py", line 88, in get_saved_model_tag_sets
    saved_model = read_saved_model(saved_model_dir)
  File
"/usr/local/lib/python3.7/dist-packages/tensorflow_core/python/tools/
saved_model_utils.py", line 55, in read_saved_model
    raise IOError("SavedModel file does not exist at: %s" %
saved_model_dir)
OSError: SavedModel file does not exist at: vgg16_saved_model
```

Inference with native TF2.0 saved model

```
model = tf.keras.models.load_model('/content/drive/MyDrive/Colab
Notebooks/vgg16_saved_model')
```

```
img_file = images[i]
img_path = os.path.join(image_dir,img_file)
img = image.load_img(img_path, target_size=(224, 224))
x = image.img_to_array(img)
x = np.expand_dims(x, axis=0)
x = preprocess_input(x)

preds = model.predict(x)
# decode the results into a list of tuples (class, description,
probability)
# (one such list for each sample in the batch)
print('{} - Predicted: {}'.format(img_path, decode_predictions(preds,
top=3)[0]))
```

```
plt.subplot(2,2,1)
plt.imshow(img);
plt.axis('off');
plt.title(decode_predictions(preds, top=3)[0][0][1])
```

```
/content/drive/MyDrive/Colab Notebooks/test/0086a6340.png - Predicted:
[('n03788365', 'mosquito_net', 0.021375598), ('n01796340',
'ptarmigan', 0.016553488), ('n07860988', 'dough', 0.012831482)]
```

```
Text(0.5, 1.0, 'mosquito_net')
```




```
batch_size = 8
batched_input = np.zeros((batch_size, 224, 224, 3), dtype=np.float32)

for i in range(batch_size):
  img_file = images[i%4]
  img_path = os.path.join(image_dir,img_file)
  img = image.load_img(img_path, target_size=(224, 224))
  x = image.img_to_array(img)
  x = np.expand_dims(x, axis=0)
  x = preprocess_input(x)
  batched_input[i, :] = x
batched_input = tf.constant(batched_input)
print('batched_input shape: ', batched_input.shape)
```

```
batched_input shape:  (8, 224, 224, 3)
```

```
# Benchmarking throughput

N_run = 300
elapsed_time = []

for i in range(N_run):
  start_time = time.time()
  preds = model.predict(batched_input)
  end_time = time.time()
  elapsed_time = np.append(elapsed_time, end_time - start_time)
  if i % 50 == 0:
    print('Step {}: {:4.1f}ms'.format(i, (elapsed_time[-50:].mean()) *
1000))
```

```
print('Throughput: {:.0f} images/s'.format(N_run * batch_size /
elapsed_time.sum()))
```

```
Step 0: 1032.0ms
Step 50: 52.6ms
Step 100: 54.2ms
Step 150: 54.1ms
Step 200: 52.6ms
Step 250: 52.6ms
Throughput: 142 images/s
```

TF-TRT FP32 model

We first convert the TF native FP32 model to a TF-TRT FP32 model.

```
print('Converting to TF-TRT FP32...')
conversion_params =
trt.DEFAULT_TRT_CONVERSION_PARAMS._replace(precision_mode=trt.TrtPreci
sionMode.FP32,

max_workspace_size_bytes=8000000000)

converter =
trt.TrtGraphConverterV2(input_saved_model_dir='/content/drive/MyDrive/
Colab Notebooks/vgg16_saved_model',

conversion_params=conversion_params)
converter.convert()
converter.save(output_saved_model_dir='vgg16_saved_model_TFTRT_FP32')
print('Done Converting to TF-TRT FP32')
```

```
Converting to TF-TRT FP32...
INFO:tensorflow:Linked TensorRT version: (5, 1, 5)
INFO:tensorflow:Loaded TensorRT version: (5, 1, 5)
INFO:tensorflow:Running against TensorRT version 5.1.5
WARNING:tensorflow:From
/usr/local/lib/python3.7/dist-packages/tensorflow_core/python/ops/
resource_variable_ops.py:1781: calling BaseResourceVariable.__init__
(from tensorflow.python.ops.resource_variable_ops) with constraint is
deprecated and will be removed in a future version.
Instructions for updating:
If using Keras pass *_constraint arguments to layers.
INFO:tensorflow:Assets written to: vgg16_saved_model_TFTRT_FP32/assets
Done Converting to TF-TRT FP32
```

```
!saved_model_cli show --all --dir vgg16_saved_model_TFTRT_FP32
```

```
MetaGraphDef with tag-set: 'serve' contains the following
SignatureDefs:
```

```
signature_def['__saved_model_init_op']:
  The given SavedModel SignatureDef contains the following input(s):
  The given SavedModel SignatureDef contains the following output(s):
    outputs['__saved_model_init_op'] tensor_info:
        dtype: DT_INVALID
        shape: unknown_rank
        name: NoOp
  Method name is:

signature_def['serving_default']:
  The given SavedModel SignatureDef contains the following input(s):
    inputs['input_1'] tensor_info:
        dtype: DT_FLOAT
        shape: (-1, 224, 224, 3)
        name: serving_default_input_1:0
  The given SavedModel SignatureDef contains the following output(s):
    outputs['predictions'] tensor_info:
        dtype: DT_FLOAT
        shape: unknown_rank
        name: PartitionedCall:0
  Method name is: tensorflow/serving/predict
```

Next, we load and test the TF-TRT FP32 model.

```
def predict_tftrt(input_saved_model):
    """Runs prediction on a single image and shows the result.
    input_saved_model (string): Name of the input model stored in the
current dir
    """
    img_file = images[0]
    img_path = os.path.join(image_dir,img_file)
    img = image.load_img(img_path, target_size=(224, 224))
    x = image.img_to_array(img)
    x = np.expand_dims(x, axis=0)
    x = preprocess_input(x)
    x = tf.constant(x)

    saved_model_loaded = tf.saved_model.load(input_saved_model,
tags=[tag_constants.SERVING])
    signature_keys = list(saved_model_loaded.signatures.keys())
    print(signature_keys)

    infer = saved_model_loaded.signatures['serving_default']
    print(infer.structured_outputs)

    labeling = infer(x)
    print(labeling.keys())
    #preds = labeling['probs'].numpy()
    preds = labeling['predictions'].numpy()
```

```
    print('{} - Predicted: {}'.format(img_path,
decode_predictions(preds, top=3)[0]))
    plt.subplot(2,2,1)
    plt.imshow(img);
    plt.axis('off');
    plt.title(decode_predictions(preds, top=3)[0][0][1])

predict_tftrt('vgg16_saved_model_TFTRT_FP32')
```

```
['serving_default']
{'predictions': TensorSpec(shape=<unknown>, dtype=tf.float32,
name='predictions')}
dict_keys(['predictions'])
Downloading data from
https://storage.googleapis.com/download.tensorflow.org/data/imagenet_c
lass_index.json
40960/35363 [==================================] - 0s 0us/step
/content/drive/MyDrive/Colab Notebooks/train/Black-grass/0050f38b3.png
- Predicted: [('n03788365', 'mosquito_net', 0.07154264), ('n15075141',
'toilet_tissue', 0.031133646), ('n04209239', 'shower_curtain',
0.022633843)]
```




```
def benchmark_tftrt(input_saved_model):
    saved_model_loaded = tf.saved_model.load(input_saved_model,
tags=[tag_constants.SERVING])
    infer = saved_model_loaded.signatures['serving_default']


    N_run = 300
    elapsed_time = []

    for i in range(N_run):
      start_time = time.time()
      labeling = infer(batched_input)
      #prob = labeling['probs'].numpy()
      end_time = time.time()
      elapsed_time = np.append(elapsed_time, end_time - start_time)
      if i % 50 == 0:
        print('Step {}: {:4.1f}ms'.format(i, (elapsed_time[-
50:].mean()) * 1000))
```

```
    print('Throughput: {:.0f} images/s'.format(N_run * batch_size /
elapsed_time.sum()))

benchmark_tftrt('vgg16_saved_model_TFTRT_FP32')
```

```
Step 0: 13852.6ms
Step 50: 47.1ms
Step 100: 48.3ms
Step 150: 48.6ms
Step 200: 48.6ms
Step 250: 48.7ms
Throughput: 85 images/s
```

TF-TRT FP16 model

We next convert the native TF FP32 saved model to TF-TRT FP16 model.

```
def predict_tftrt(input_saved_model):
    """Runs prediction on a single image and shows the result.
    input_saved_model (string): Name of the input model stored in the
current dir
    """
    img_file = images[0]
    img_path = os.path.join(image_dir,img_file)
    img = image.load_img(img_path, target_size=(224, 224))
    x = image.img_to_array(img)
    x = np.expand_dims(x, axis=0)
    x = preprocess_input(x)
    x = tf.constant(x)

    saved_model_loaded = tf.saved_model.load(input_saved_model,
tags=[tag_constants.SERVING])
    signature_keys = list(saved_model_loaded.signatures.keys())
    print(signature_keys)

    infer = saved_model_loaded.signatures['serving_default']
    print(infer.structured_outputs)

    labeling = infer(x)
    preds = labeling['predictions'].numpy()
    print('{} - Predicted: {}'.format(img_path,
decode_predictions(preds, top=3)[0]))
    plt.subplot(2,2,1)
    plt.imshow(img);
    plt.axis('off');
    plt.title(decode_predictions(preds, top=3)[0][0][1])

print('Converting to TF-TRT FP16...')
conversion_params = trt.DEFAULT_TRT_CONVERSION_PARAMS._replace(
    precision_mode=trt.TrtPrecisionMode.FP16,
```

```
    max_workspace_size_bytes=8000000000)
converter = trt.TrtGraphConverterV2(
   input_saved_model_dir='/content/drive/MyDrive/Colab
Notebooks/vgg16_saved_model', conversion_params=conversion_params)
converter.convert()
converter.save(output_saved_model_dir='vgg16_saved_model_TFTRT_FP16')
print('Done Converting to TF-TRT FP16')
```

```
Converting to TF-TRT FP16...
INFO:tensorflow:Linked TensorRT version: (5, 1, 5)
INFO:tensorflow:Loaded TensorRT version: (5, 1, 5)
INFO:tensorflow:Running against TensorRT version 5.1.5
WARNING:tensorflow:From
/usr/local/lib/python3.7/dist-packages/tensorflow_core/python/ops/
resource_variable_ops.py:1781: calling BaseResourceVariable.__init__
(from tensorflow.python.ops.resource_variable_ops) with constraint is
deprecated and will be removed in a future version.
Instructions for updating:
If using Keras pass *_constraint arguments to layers.
INFO:tensorflow:Assets written to: vgg16_saved_model_TFTRT_FP16/assets
Done Converting to TF-TRT FP16
```

```
predict_tftrt('vgg16_saved_model_TFTRT_FP16')
```

```
['serving_default']
{'predictions': TensorSpec(shape=<unknown>, dtype=tf.float32,
name='predictions')}
/content/drive/MyDrive/Colab Notebooks/train/Black-grass/0050f38b3.png
- Predicted: [('n03788365', 'mosquito_net', 0.07154888), ('n15075141',
'toilet_tissue', 0.03112983), ('n04209239', 'shower_curtain',
0.022629205)]
```




```
def benchmark_tftrt(input_saved_model):
    saved_model_loaded = tf.saved_model.load(input_saved_model,
tags=[tag_constants.SERVING])
    infer = saved_model_loaded.signatures['serving_default']


    N_run = 300
    elapsed_time = []
```

```
    for i in range(N_run):
      start_time = time.time()
      labeling = infer(batched_input)
      #prob = labeling['probs'].numpy()
      end_time = time.time()
      elapsed_time = np.append(elapsed_time, end_time - start_time)
      if i % 50 == 0:
        print('Step {}: {:4.1f}ms'.format(i, (elapsed_time[-
50:].mean()) * 1000))

    print('Throughput: {:.0f} images/s'.format(N_run * batch_size /
elapsed_time.sum()))

benchmark_tftrt('vgg16_saved_model_TFTRT_FP16')
```

```
Step 0: 4364.5ms
Step 50: 49.7ms
Step 100: 48.9ms
Step 150: 49.0ms
Step 200: 49.2ms
Step 250: 49.3ms
Throughput: 126 images/s
```

TF-TRT INT8 model

Creating TF-TRT INT8 model requires a small calibration dataset. This data set ideally should represent the test data in production well, and will be used to create a value histogram for each layer in the neural network for effective 8-bit quantization.

Herein, for demonstration purposes, we take only the 4 images that we downloaded for calibration. In production, this set should be more representative of the production data.

Due to Colab memory limit which sometime causes TensorRT to crash, to proceed, first, restart the runtime by pressing CTRL+M or select *Runtime -> Restart runtime...* or simply execute the next cell.

```
# import os
# os.kill(os.getpid(), 9)
```

```
batch_size = 8
batched_input = np.zeros((batch_size, 224, 224, 3), dtype=np.float32)

for i in range(batch_size):
  img_file = images[i%4]
  img_path = os.path.join(image_dir,img_file)
  img = image.load_img(img_path, target_size=(224, 224))
  x = image.img_to_array(img)
  x = np.expand_dims(x, axis=0)
  x = preprocess_input(x)
  batched_input[i, :] = x
```

```
batched_input = tf.constant(batched_input)
print('batched_input shape: ', batched_input.shape)
```

```
batched_input shape:  (8, 224, 224, 3)
```

```
print('Converting to TF-TRT INT8...')
conversion_params = trt.DEFAULT_TRT_CONVERSION_PARAMS._replace(
    precision_mode=trt.TrtPrecisionMode.INT8,
    max_workspace_size_bytes=8000000000,
    use_calibration=True)
converter = trt.TrtGraphConverterV2(
    input_saved_model_dir='/content/drive/MyDrive/Colab
Notebooks/vgg16_saved_model',
    conversion_params=conversion_params)

def calibration_input_fn():
    yield (batched_input, )
converter.convert(calibration_input_fn=calibration_input_fn)

converter.save(output_saved_model_dir='vgg16_saved_model_TFTRT_INT8')
print('Done Converting to TF-TRT INT8')
```

```
Converting to TF-TRT INT8...
INFO:tensorflow:Linked TensorRT version: (5, 1, 5)
INFO:tensorflow:Loaded TensorRT version: (5, 1, 5)
INFO:tensorflow:Running against TensorRT version 5.1.5
```

```
def predict_tftrt(input_saved_model):
    """Runs prediction on a single image and shows the result.
    input_saved_model (string): Name of the input model stored in the
current dir
    """
    img_file = images[0]
    img_path = os.path.join(image_dir,img_file)
    img = image.load_img(img_path, target_size=(224, 224))
    x = image.img_to_array(img)
    x = np.expand_dims(x, axis=0)
    x = preprocess_input(x)
    x = tf.constant(x)

    saved_model_loaded = tf.saved_model.load(input_saved_model,
tags=[tag_constants.SERVING])
    signature_keys = list(saved_model_loaded.signatures.keys())
    print(signature_keys)

    infer = saved_model_loaded.signatures['serving_default']
    print(infer.structured_outputs)

    labeling = infer(x)
    preds = labeling['probs'].numpy()
    print('{} - Predicted: {}'.format(img_path,
```

```
decode_predictions(preds, top=3)[0]))
    plt.subplot(2,2,1)
    plt.imshow(img);
    plt.axis('off');
    plt.title(decode_predictions(preds, top=3)[0][0][1])

predict_tftrt('vgg16_saved_model_TFTRT_INT8')
```

```
['serving_default']
{'probs': TensorSpec(shape=<unknown>, dtype=tf.float32, name='probs')}
/content/drive/MyDrive/Colab Notebooks/train/Black-grass/0050f38b3.png
- Predicted: [('n03930313', 'picket_fence', 0.21861497), ('n03089624',
'confectionery', 0.21403234), ('n07880968', 'burrito', 0.098421164)]
```




```
def benchmark_tftrt(input_saved_model):
    saved_model_loaded = tf.saved_model.load(input_saved_model,
tags=[tag_constants.SERVING])
    infer = saved_model_loaded.signatures['serving_default']

    N_run = 300
    elapsed_time = []

    for i in range(N_run):
      start_time = time.time()
      labeling = infer(batched_input)
      #prob = labeling['probs'].numpy()
      end_time = time.time()
      elapsed_time = np.append(elapsed_time, end_time - start_time)
      if i % 50 == 0:
        print('Step {}: {:4.1f}ms'.format(i, (elapsed_time[-
50:].mean()) * 1000))

    print('Throughput: {:.0f} images/s'.format(N_run * batch_size /
elapsed_time.sum()))

benchmark_tftrt('vgg16_saved_model_TFTRT_INT8')
```

```
WARNING:tensorflow:Unresolved object in checkpoint:
(root).trt_engine_resources.TRTEngineOp_0._serialized_trt_resource_fil
ename
```

```
WARNING:tensorflow:A checkpoint was restored (e.g.
tf.train.Checkpoint.restore or tf.keras.Model.load_weights) but not
all checkpointed values were used. See above for specific issues. Use
expect_partial() on the load status object, e.g.
tf.train.Checkpoint.restore(...).expect_partial(), to silence these
warnings, or use assert_consumed() to make the check explicit. See
https://www.tensorflow.org/alpha/guide/checkpoints#loading_mechanics
for details.
Step 0: 1442.1ms
Step 50:  5.8ms
Step 100:  3.4ms
Step 150:  3.4ms
Step 200:  3.3ms
Step 250:  3.3ms
Throughput: 934 images/s
```

Conclusion

In this notebook, we have demonstrated the process of creating TF-TRT FP32, FP16 and INT8 inference models from an original Keras FP32 model, as well as verify their speed and accuracy.

What's next

Try TF-TRT on your own model and data, and experience the simplicity and speed up it offers.